Signal Processing of the Inclination Meter from DDS Excitation to Displacement Readout

Draft — work in progress (reflects firmware up to 0.10.70, 2026-10-06)

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1 Scope and Conventions

This note describes the measurement signal chain of the REV B instrument, from the sine wave generated by the DDS to the displacement value delivered by the firmware. It aims to be unambiguous: every quantity has a name, a unit, a sign convention and a place in the chain. The chain is developed one stage at a time. This part covers the *excitation*: the DDS up to the two antiphase voltages A and B at the op-amp outputs, and how these are attenuated, filtered and fed into the ADC. It then introduces the Wyler Leveltronic sensor, treated as a black box, and how its output is filtered and fed into the ADC. The remaining sections cover the sampling, the amplitude and phase estimate, the complex ratio, and the processing that follows it, and end with a list of open items and proposals.

Sources. Component values come from the KiCad schematic (`SignalProcessing.kicad_sch`), *except* where the built REV B boards differ from it; such deviations are stated explicitly (currently R52, R46–R49). Device figures come from the AD9833, AD826 and ADS131M04 datasheets. Component tolerances are not recorded in the schematic and are not used here.

Conventions. All AC signals are sinusoids at the excitation frequency f_0. Amplitudes are *peak* values, not RMS. Phases are quoted relative to the DDS output; a positive phase means the signal leads. Later parts re-reference phases to the ADC sampling grid.

2 Overview

Stage	Parts	Function	Output
DDS	U14 (AD9833)	10-bit sine, $f_0 \approx 2604\,\text{Hz}$	—
Filter	C59, R50, R51, C60, C61	DC block + RC low-pass ($R_{52} = 0$)	`SineSignal`
Gain stage	U16A, R53, R54	non-inverting, $G_1 = +11$	`ExciterA`
Inverter	U16B, R55, R56	inverting, $G_2 = -1$	`ExciterB`
A/B input network	R36, R38, R40, C53, R44, R48, C57 (A); R37, R39, R41, C54, R45, R49, C58 (B)	attenuator, AC coupling, RC filter	ADC pins AIN2P (A), AIN1P (B)
Sensor input network	C51, R42, D5–D8, R46, C55 (S1); C52, R43, D9–D12, R47, C56 (S2)	ground reference, clamps, RC filter	ADC pins AIN3P (S1), AIN0P (S2)

The op-amp U16 (AD826, dual) runs from 5 V and -4.5 V rails; the reference for all AC signals is ground.

3 Sine Generation (DDS)

The AD9833 is clocked by a timer output (TIM1 channel 4) at $f_{\text{MCLK}} = 64\,\text{MHz}/12 = 5.3333\,\text{MHz}$ and programmed once at boot with FREQREG $= 2^{17}$. Exactly 2048 MCLK cycles make one period:

$$\boxed{f_0 = f_{\text{MCLK}}/2^{11} = 2604.17\,\text{Hz}.} \tag{1}$$

The output pin is a $200\,\Omega$ source delivering a *unipolar* sine. From the datasheet limits (38 mV to 650 mV) it has a DC offset of 344 mV and a peak amplitude

$$V_d = 306\,\text{mV}. \tag{2}$$

These are typical values with no guaranteed tolerance, and `VOUT` drifts by about $200\,\text{ppm}/^\circ\text{C}$. The absolute amplitude is therefore *not* a calibrated quantity; the measurement is designed so that it cancels (ratio stage, later part).

The DAC is 10 bit and updates at f_{MCLK}, so the output is a staircase with spectral images at $kf_{\text{MCLK}} \pm f_0$. Removing them is one job of the filter.

4 Passive Filter

R52 is bridged on REV B. The schematic shows R52 as $68\,\text{k}\Omega$, which would make a two-stage RC filter. On the built boards it is $0\,\Omega$, because the $68\,\text{k}\Omega$ sat in the op-amp's bias-current path and caused a large DC offset (section 6). This is a REV B hardware workaround. With it, C60 and C61 are in parallel, $C_p = 1.1\,\text{nF}$, and the filter has two parts:

- **DC block** (C59, R50): removes the DDS offset; high-pass corner about 156 Hz.
- **RC low-pass** (R51, C_p): a single first-order stage, $f_{\text{LP}} = 1/(2\pi R_{51} C_p) = 21.3\,\text{kHz}$.

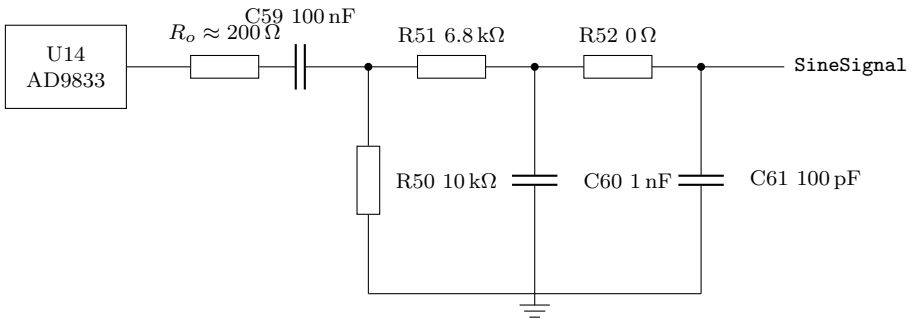


Figure 1: Filter between the DDS and the op-amp, as built on REV B. The schematic file still shows $R_{52} = 68\,\text{k}\Omega$; on the boards it is bridged ($0\,\Omega$), so C60 and C61 are in parallel.

Solving the network exactly (including the $200\,\Omega$ source and the loading of R50, R51 and C_p; the op-amp input is treated as an open circuit) gives, at f_0:

	Magnitude	Phase
DC block	0.968	3.3°
Low-pass	0.993	−7.0°
Filter total	**0.961**	−3.7°

Above the corner the roll-off is first order. At f_{MCLK} the filter attenuates by only about 48 dB (the original two-stage design gave about 95 dB), so the DAC images reach the op-amp only mildly suppressed. Whether this matters depends on the ADC's own filtering and is treated in the acquisition part.

5 Gain Stage and Inverter

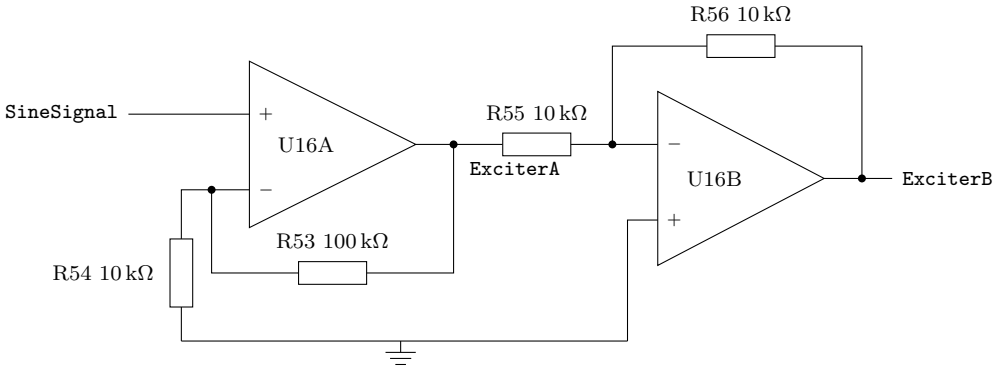


Figure 2: The two op-amp stages (U16A non-inverting, U16B inverting). Supplies: `5V0` on pin 8, `-4V5` on pin 4.

U16A (non-inverting). The filtered sine enters the non-inverting input; R53 and R54 set

$$G_1 = 1 + R_{53}/R_{54} = 11. \tag{3}$$

The output is the net `ExciterA`. The AD826 (50 MHz gain-bandwidth product) is far faster than needed: at f_0 its phase error is about 0.03°, which is ignored.

U16B (inverting). R55 and R56 (both 10 kΩ) make a unity-gain inverter, $G_2 = -R_{56}/R_{55} = -1$, whose output is `ExciterB`. So B has the same amplitude as A and is exactly 180° out of phase, up to the ratio R_{56}/R_{55}. Any common change (DDS amplitude, filter, G_1) scales A and B

identically; only the resistor ratio makes them differ. The ratio calculation in the digital part relies on this.

Result.

$$\boxed{|V_A| = |V_B| = G_1 \cdot 0.961 \cdot V_d = 11 \cdot 0.961 \cdot 0.306\,\mathrm{V} \approx 3.23\,\mathrm{V}\ \text{(peak)},} \tag{4}$$

with phases $\varphi_A = -3.7°$ and $\varphi_B = 176.3°$ relative to the DDS output.

6 DC Operating Point

The AD826 has bipolar inputs drawing a bias current of 3.3 µA (typical, 6.6 µA maximum). The capacitors block DC, so this current must flow to ground through R50, R51 and R52. The voltage it develops across that resistance is amplified by G_1 and appears as a DC offset on `ExciterA` (and, inverted, on `ExciterB`):

	DC path resistance	Offset, 3.3 µA	Offset, 6.6 µA
Schematic, $R_{52} = 68\,\mathrm{k\Omega}$	84.8 kΩ	−2.75 V	−5.5 V
REV B, $R_{52} = 0\,\Omega$	16.8 kΩ	−0.28 V	−0.56 V

With the schematic value, the 3.23 V AC peak on a −2.75 V offset would need about −6 V, beyond the −4.5 V rail; this was seen as a DC offset problem on the board and is why R52 was bridged.

With $R_{52} = 0$ the residual offset is small, but headroom on the negative peak of `ExciterA` (positive peak of `ExciterB`) is still thin: about 3.5 V are needed, against an AD826 output swing of ±3.3 V (guaranteed) to ±3.8 V (typical). **Status:** reported to work; DC level and peak values on the two nets have not been measured yet and should be recorded here once they are.

7 Attenuator and ADC Input Filter (Channels A and B)

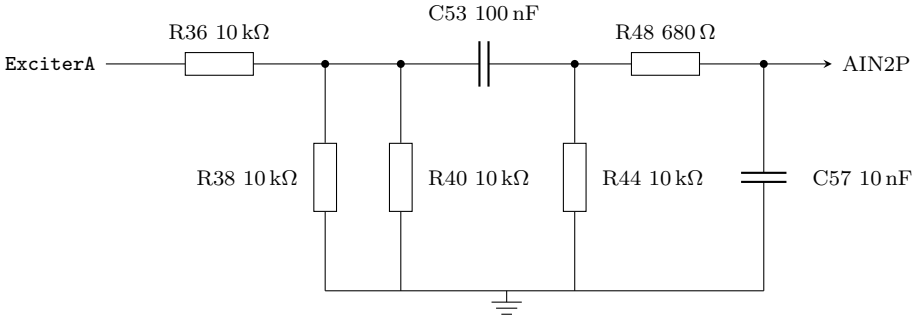


Figure 3: Input network of channel A, as built on REV B (R48 is 680 Ω, not the 680 kΩ of the schematic). Channel B is identical (R37, R39, R41, C54, R45, R49, C58) and ends at AIN1P. The ADC's negative inputs are tied to ground.

REV B deviation. The schematic and the manufactured boards specify R48 and R49 as 680 kΩ. This was a mistake: with 10 nF it gives a corner of 23 Hz and would leave almost none of the 2.6 kHz signal at the ADC. The working boards carry 680 Ω instead (hardware rework), which is what is described here.

What the parts do.

- **R36 with R38‖R40** (10 kΩ over 5 kΩ): attenuator, ratio 1/3. It brings the 3.2 V peak of the exciter into the ADC's ±1.2 V range. (Since fw 0.10.67 this attenuator has no constant of its own in the firmware: it is part of the empirical $|k|$, Section 13.3.)

- **C53 with R44:** AC coupling. C53 removes the DC offset of the exciter (section 6); R44 gives the node a DC path to ground. R44 also loads the attenuator, so the total division is larger than 1/3.
- **R48 with C57:** RC low-pass directly at the ADC pin, with an isolated corner of $1/(2\pi \cdot 680\,\Omega \cdot 10\,\text{nF}) = 23.4\,\text{kHz}$.

The stages are not independent. Since R48 is small, C57 is not isolated from R44: the node after C53 sees R44 in parallel with R48+C57. The effective low-pass corner is therefore set mostly by the source resistance in front of C57 ($R_{36} \parallel R_{38} \parallel R_{40} \parallel R_{44} \approx 2.5\,\text{k}\Omega$, plus R48), giving about 5 kHz, not 23 kHz. Solving the network exactly (ADC input treated as an open circuit) gives a band-pass response with its maximum (gain 0.231) near 0.8 kHz and its -3 dB points at 108 Hz and 5.5 kHz. At the excitation frequency, on the upper skirt of that band:

$$\boxed{H_{AB}(\text{j}\omega_0) = 0.212 \angle -23.7^\circ \quad (-13.5\,\text{dB}).} \tag{5}$$

At f_{MCLK} the network attenuates by a further 73 dB (on top of the 48 dB of the first filter). Channels A and B use identical networks, so this gain and phase are common to both. They matter as soon as A or B is compared with a differently filtered channel, such as the sensor inputs.

ADC input. Each channel is single-ended: the positive input sees the network output, the negative input is grounded. The PGA gain of the A and B channels is 1, so the full-scale range is ± 1.2 V and one code (LSB) is $2.4\,\text{V}/2^{24} = 143\,\text{nV}$. The ADC's input impedance at gain 1 is 330 kΩ at a modulator frequency of 4.096 MHz and scales inversely with it, so about 510 kΩ here (Section 10). This is large against R48; including it changes the result by less than 1 % and it is ignored. The expected amplitude at the ADC pin is

$$|V_{A,\text{pin}}| = |V_{B,\text{pin}}| = 0.212 \cdot 3.23\,\text{V} \approx 0.68\,\text{V (peak)}, \tag{6}$$

about 57 % of full scale, leaving a factor of 1.75 headroom.

Cross-check with the firmware. The firmware reports about 885 mV RMS for channel A, i.e. 1.25 V peak. That would exceed the ± 1.2 V range, and yet no clipping is counted. The reason is a unit factor: the firmware diagnostic (`phasor_peak_mv` in `svc_displacement.c`) converts codes with $2400/\text{PGA}/2^{23}$, twice the datasheet LSB. With the datasheet LSB the measured peak is about 0.63 V, 9 % below the prediction above — consistent within the unspecified DDS amplitude tolerance (the prediction uses the typical DDS amplitude). **Confirmed and fixed (2026-10-06):** the full scale of the ADC is $\pm 1.2\,\text{V/PGA}$ for $\pm 2^{23}$ codes, so one LSB is $2.4\,\text{V}/(\text{PGA} \cdot 2^{24})$; the firmware used 2^{23} in the denominator of the 2.4 V span. `phasor_peak_mv` now uses $1200/\text{PGA}/2^{23}$ (the same constant in the Python test tools was corrected too). The error only affected the diagnostic amplitudes (the DIAGNOSTICS screen, the API signal diagnostics) and the "theoretical tilt" derived from them; the displacement readings do not use it.

8 The Sensor: Wyler Leveltronic A40 (Black Box)

The sensor is a Wyler Leveltronic A40 inclination sensor in its 5 µm/m-per-digit version (source: the Wyler operating manual, 06/87). It is treated here as a black box; only its electrical behaviour at the terminals matters.

Principle, in brief. A mass disc of about 1 g hangs on leaf springs between two electrodes, forming a differential capacitor; air damping and the friction-free suspension give the good repeatability. Tilting the instrument moves the disc toward one electrode and away from the other. The original electronics (Levelmeter 25) is a carrier-frequency system (2.5 kHz) with phase-sensitive rectification. In this instrument that job is done by the ADC and the digital processing instead.

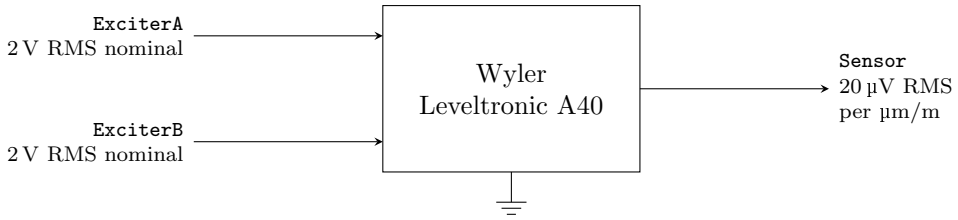


Figure 4: The sensor as a black box: two antiphase supply inputs and one signal output, plus ground.

8.1 Electrical behaviour

- **Powered by the exciter signals.** The sensor has no other supply. It is fed directly with the two antiphase excitation voltages A and B (2 V RMS at 2.5 kHz nominal in the manual). Their generation is described in the sections above.
- **The output is a sine wave** at the excitation frequency. Its amplitude is proportional to the magnitude of the tilt.
- **The sign of the tilt is the phase of the output relative to A and B.** For one sign the output is in phase with A (and so in antiphase with B); for the other sign it is in phase with B. The output passes through zero amplitude at zero tilt.
- **Sensitivity:** 20 µV RMS per 0.001 mm/m (that is, per 1 µm/m) for the 5 µm/m-per-digit version, which is the one used here. This agrees with the manual's 100 µV RMS per digit of 5 µm/m.

8.2 Model

Let τ be the tilt in µm/m, positive when the output is in phase with A, and let $\underline{V}_S$ be the phasor of the sensor output. Since $\underline{V}_B = -\underline{V}_A$, the behaviour above is described by a single real number per unit tilt:

$$\underline{V}_S = \kappa\,\tau\,\underline{V}_A, \qquad \kappa \approx \frac{20\,\mu\text{V}}{2\,\text{V}} = 10^{-5} \text{ per µm/m}. \tag{7}$$

A positive τ gives $\underline{V}_S$ in phase with $\underline{V}_A$, a negative τ in phase with $\underline{V}_B$. Equation (7) also shows that the output is *ratiometric*: it scales with the excitation amplitude, so the ratio $\underline{V}_S/\underline{V}_A$ measures the tilt independent of that amplitude. (This is an inference from the sensor being supplied by the exciter, with the nominal figures taken at 2 V.)

Three points on how this maps to the present instrument:

- The excitation here is 2.29 V RMS at 2.604 kHz, i.e. 14 % above the nominal supply and 4 % above the nominal carrier frequency. With (7) the expected sensitivity is about 23 µV RMS per µm/m at the sensor output, to be verified.
- The sign convention $\tau > 0 \Leftrightarrow$ output in phase with A is the *electrical* one. Which physical direction is positive depends on how the instrument is mounted and is fixed per instrument during calibration.
- The manual specifies a range of ±1999 digits, i.e. ±10 mm/m for this version, which corresponds to about 200 mV RMS at the sensor output (at nominal excitation).

How the sensor output reaches the ADC is described in the next section.

9 Sensor Input Network and ADC (Channels S1 and S2)

Two sensors can be connected. The output of sensor 1 (net `Sensor1`, connector J5) goes through the network of Figure 5 to ADC input AIN3P (channel 3); sensor 2 (`Sensor2`, J6) has an identical network (C52, R43, D9–D12, R47, C56) into AIN0P (channel 0). As for A and B, the ADC's negative inputs are grounded.

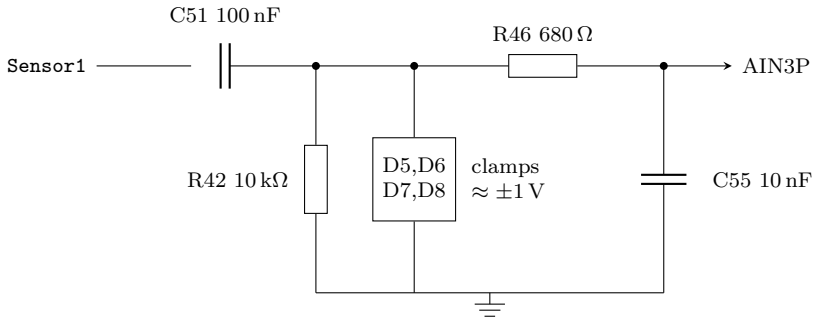


Figure 5: Input network of sensor 1, as built on REV B (R46 is 680 Ω, not the 680 kΩ of the schematic). Each clamp block is two series diodes (1N4148WS) each way: D5, D6 to ground for negative excursions, D8, D7 for positive ones.

REV B deviation. As for R48 and R49, the schematic and the manufactured boards specify R46 and R47 as 680 kΩ, which was a mistake (a 23 Hz corner with 10 nF). The working boards carry 680 Ω.

What the parts do.

- **C51 with R42: ground reference.** The sensor signal is AC; C51 blocks any DC and R42 ties the node to ground, so the signal is centred on 0 V at the ADC. Together they form a high-pass with corner $1/(2\pi \cdot 100\,\text{nF} \cdot 10\,\text{k}\Omega) = 159\,\text{Hz}$.
- **D5–D8: protection clamps.** Two diode drops in each direction limit the node to roughly ±1 V, below the ADC's absolute input limit. In normal operation the signal is a few tens of millivolts and the diodes do not conduct.
- **R46 with C55: RC low-pass** at the ADC pin, isolated corner $1/(2\pi \cdot 680\,\Omega \cdot 10\,\text{nF}) = 23.4\,\text{kHz}$.

Transfer function. Solving the network exactly, with the sensor treated as an ideal low-impedance source and the ADC input as an open circuit (its input impedance is at least 1 MΩ at gain 16), gives a band-pass with maximum gain 0.904 near 1.9 kHz and −3 dB points at 143 Hz and 26 kHz. At the excitation frequency

$$\boxed{H_S(\mathrm{j}\omega_0) = 0.903 \angle -2.6^\circ.} \tag{8}$$

The response at f_{MCLK} is −47 dB. This network is much wider and much less attenuating than that of A and B (Section 7), because no attenuator sits in front of it. Two consequences:

- **Gain:** $|H_S| = 0.903$ holds only if the sensor output impedance is small. A series source resistance of 1 kΩ already lowers it to 0.81. The sensor's output impedance is not specified in the manual; the output is an op-amp output (low impedance), so the assumption is reasonable but not measured.
- **Phase:** the S network lags by 2.6°, the A/B network by 23.7°. In the ADC domain the sensor signal therefore carries a phase offset of 21° relative to A and B, on top of anything the sensor itself adds. The ratio calculation compares complex phasors, so this offset has to be accounted for in the digital processing.

ADC input. Channels 3 and 0 run with a PGA gain of 16 (A and B: 1). The full-scale range is therefore ±75 mV and one LSB is $2.4\,\mathrm{V}/(16 \cdot 2^{24}) = 8.9\,\mathrm{nV}$. With the expected 23 µV RMS per µm/m at the sensor output (Section 8) and $|H_S|$, the sensitivity at the ADC pin is

$$\approx 21\,\mu\mathrm{V\ RMS\ per\ \mu m/m}, \tag{9}$$

so full scale corresponds to about ±2.5 mm/m tilt. This is a quarter of the sensor's own ±10 mm/m range; larger tilts need a lower PGA gain or would clip.

Observed phase. The long granite-plate run (both sensors with a large signal) shows the following average phase differences in the ADC domain. They are given as the *delay* of the first signal behind the second (positive = later); this is the angle the firmware's demodulator reports (Section 11).

Quantity	Measured
B behind A	178.9° (ideal: 180°)
S_1 behind B	−10.0°
S_2 behind B	−13.8°

Both sensor signals were in phase with B (negative tilt) to within about 10°–14°. The networks alone delay the S signal by 2.6° and the A/B signals by 23.7°, so the S signals should arrive 21° *earlier* than B (−21°). Measured are −10° and −14°, so about 11° (S_1) and 7° (S_2) of additional delay remain that must come from the sensor itself: its output is a low-impedance op-amp output, so the extra delay comes from the old electronics inside the sensor, not from loading. **Open:** the delay needs to be characterised; also the 1° deviation of B from exactly 180° relative to A is unexplained (candidates: resistor ratio, channel-to-channel timing in the ADC). It is, however, *stable*: in three separate runs it came out 178.94°, 178.96° and 179.00° (the absolute phase of A in the same runs was not stable, Section 10.5), so it behaves as a fixed property of the measurement paths.

9.1 Phase calibration

Status: approach 2 below is implemented in fw 0.10.66 (see "Implementation" at the end of this subsection). The default constants are estimates and the proper calibration is still to be done.

Two approaches were considered (Table 1). **Decision: approach 2.** Its decisive advantage is that it does not rely on a property that was measured not to hold (a reproducible absolute phase), it needs fewer constants, and it costs nothing extra since $\underline{D}$ is the denominator of the ratio anyway. Approach 1 would only become attractive if the sample-grid alignment were made reproducible by a firmware change, which has no other benefit. The remainder of this subsection describes approach 2.

In theory A and B are exactly 180° apart, and S_1 and S_2 are either in phase with A or in phase with B, depending on the direction of tilt. In the ADC domain this is distorted by the input filters (different for the A/B and the S paths), by the sensor itself and by further effects such as channel timing. The measured values above show the size of the distortion. The intended remedy is an *empirical* calibration, with the following design decisions.

- **A and B are rotated by the same amount.** Their phase difference is not calibrated away. The measured 178.9° (ideal 180°) is not the inverter's delay: as a unity-gain inverter the AD826 (closed-loop bandwidth about 25 MHz) delays B by only about 0.006° at f_0, and in the wrong direction (the measured B is 1.1° *earlier* than an ideal inversion). The difference is therefore more likely due to the measurement paths (tolerances of the two input networks, ADC channel timing), but whether it is real or not, it has almost no effect on the

	Approach 1: fixed absolute constants	Approach 2: run-time reference $\underline{D}$
Idea	Store one common phase constant for A and B, plus one each for S_1 and S_2, so that A lands at 90° and B at 270° in the ADC frame.	Measure $\underline{D} = \underline{A} - \underline{B}$ in every batch and rotate all phasors so that it sits at 90°; store only the phase of each sensor relative to $\underline{D}$ (δ_1, δ_2).
Constants	3 (common A/B, S_1, S_2)	2 (δ_1, δ_2)
Needs a reproducible absolute phase	**Yes.** The constants are only valid if the sample-grid alignment is the same at every start.	No: the rotation is taken from the measured $\underline{D}$, so any grid alignment drops out.
Observed	The absolute phase of A was 122.85°, 122.67° and 334.39° in three runs (Section 10.5): not reproducible.	With $\underline{D}$ at 90°, A read 90.50–90.53° and B 269.47–269.50° in all three runs.
Drift of the common A/B path	Not followed (a fixed constant).	Followed automatically; only the *sensor-versus-excitation* phase remains calibrated.
Calibration effort	Needs a calibration after every grid-alignment change, or a firmware change that makes the alignment reproducible.	Once per sensor and after any rework of the input networks.
Cost	Simple arithmetic.	One complex division per batch for $\underline{D}$, which the ratio already computes.

Table 1: Phase calibration: fixed absolute constants against a run-time reference.

tilt: it appears as a static imaginary part of the balance term $\underline{z}$ ($\mathrm{Im}\,\underline{z} \approx \delta/4 = 4.7 \cdot 10^{-3}$ for $\delta = 1.07°$, matching the logged $-4.65 \cdot 10^{-3}$) and enters the real part used for the reading only in second order.

- **The reference is the difference phasor** $\underline{D} = \underline{A} - \underline{B}$, the denominator of the ratio. It is the excitation the sensor actually sees between its two supply inputs and lies half way between A and $-B$. Rotating all phasors so that $\underline{D}$ is real makes the projection $\mathrm{Re}[\underline{S}/(k\underline{D})]$ exactly the in-phase component. The rotation is determined at run time from the measured $\underline{D}$, which removes any dependence on the absolute phase of the sample grid (Section 10.5).

- **The reference angle is 90°, not 0°.** This changes nothing in the calculation, but the firmware's angle is $90° - \varphi$ (Section 11), so 90° is exactly a sine with zero phase at the first sample of a period, the natural zero of the convention used here. It also keeps all values away from the 0°/360° wrap: with $\underline{D}$ at 90°, A reads about 90.5°, B about 269.5° (symmetric, because $\underline{D}$ is the midpoint), and each sensor lies near 90° (positive tilt, in phase with A) or near 270° (negative tilt, in phase with B), instead of one value fluctuating around 0° and jumping between 359° and 1°. Near zero tilt the amplitude of $\underline{S}$ vanishes, its phase is undefined and jumps between the two values whatever the reference; the sign of the reading comes from the projection, not from this angle, and the angle should only be displayed above an amplitude threshold.

- **Constants:** only the phase of each sensor relative to $\underline{D}$, δ_1 and δ_2, needs to be stored in EEPROM. This is the phase of the complex constant $k = |k|e^{\mathrm{j}\delta}$ of Section 12 (one per sensor). No individual constants for A and B are needed.

- **Determination:** apply a clearly non-zero tilt in a known direction and choose δ so that $\underline{S}$ lands on the real axis with the sign given by that direction. The best estimate uses the *change* of $\underline{S}$ between two clearly different tilt states (a deliberate tilt step of at least about 0.5 mm/m, *not* the zero calibration, which is done almost flat and is kept separate): the direction of the change $\Delta\underline{S}$ is the direction of $\underline{k}\,\underline{D}$ and is independent of any static offset (the quadrature zero offset and the leakage of the sensor midpoint, which are of

order 10 to 30 µm/m equivalent). Using the phase of $\underline{S}$ itself, as was done first, mixes in these offsets. From the 24 h run (quiet part, $\underline{D}$ at 90°) both sensors were in phase with B but 10.5° (S_1, reading 259.5°) and 14.3° (S_2, reading 255.7°) earlier than the ideal 270°; the later contiguous phasor stream gives about −10° for S_1 and −10.6° to −14.3° for S_2 (the range depends on the estimate used). These are estimates of δ_1 and δ_2 only; a proper determination from a controlled tilt change is still to do.

How much does it matter? Less than one might expect. The complex ratio is already invariant against a common rotation of all phasors, so the rotation of the reference to 90° changes the display only, not the result; only δ_1 and δ_2 change the arithmetic. With a phase error δ of the sensor signal the in-phase reading is reduced by $\cos\delta$: 1.7 % for 10.5° and 3.1 % for 14.3°, a scale factor that the empirical sensitivity calibration against a known tilt absorbs anyway. The other effect is that a quadrature disturbance leaks into the reading with the factor $\sin\delta$ (0.18 and 0.25). In the 24 h data the quadrature part of $\underline{S}/\underline{D}$ is however 6 to 7 times quieter than the in-phase part (Allan deviation at 1 min, in equivalent µm: S_1 0.03 against 0.28, S_2 0.07 against 0.46), so this leakage is negligible (about 0.005 µm). The real benefits are therefore a clean diagnostic (the residual becomes a true quadrature signal, the phase display is unambiguous) and protection against a scale drift when δ changes (≈ 0.3 to 0.4 % per degree), which matters for large tilts (about 7 µm/m per degree at 2.5 mm/m) but not near zero. When the signal is large (for example above 0.5 mm/m) the angle of the residual gives a free online estimate of δ, which can be used to monitor the stored constants. The measure of priority is correspondingly lower than that of the imbalance correction (Section 12.7), which reduced the drift by a factor of 4 to 8.

The correction is applied after demodulation, so it does not matter where in the analog chain each phase shift arises. The constants are valid as long as the filters and the sensor do not change; they must be re-determined after any rework of the input networks, or if the phases turn out to drift with temperature.

Implementation (fw 0.10.66). Per batch the firmware forms, as before, $\underline{u} = \underline{S}/(|k|\,\underline{D})$ with the real magnitude $|k|$ and the shared reciprocal of $\underline{D}$, and then rotates it by $-\delta$:

$$x_{\mathrm{re}} = u_{\mathrm{re}}\cos\delta + u_{\mathrm{im}}\sin\delta, \qquad x_{\mathrm{im}} = -u_{\mathrm{re}}\sin\delta + u_{\mathrm{im}}\cos\delta, \tag{10}$$

with the tilt = $(x_{\mathrm{re}} - \mathrm{zero})/k$ (Eq. (22)) and the residual x_{im}. A sensor delayed by δ relative to the ideal has positive δ, an earlier one negative (the firmware's angle convention, Section 11.1); both sensors are early, so both δ are negative. The cost is one extra complex multiplication per sensor and batch ($\cos\delta$, $\sin\delta$ are cached and only recomputed when the constant changes). Nothing is needed for A and B: $\underline{D}$ is measured in every batch, so the sample-grid phase drops out of the reading. The diagnostic phases (DIAGNOSTICS screen, API signal diagnostics) are shown relative to $\underline{D}$ at 90°, in the range 0 to 360°.

Item	Value
Constants	`disp_s1_phase_cdeg`, `disp_s2_phase_cdeg` (int16, centidegrees, ±180°; beyond ±90° equals the other sign with the invert flag toggled), EEPROM displacement page
API	Calibrations, resources 0x42 (S_1) and 0x46 (S_2) since API v3 (0x0D/0x0E before), GET/SET
Defaults	−6.00° (S_1), −9.15° (S_2)

Defaults and check against the 19 h phasor stream. The defaults are the principal axis of $\underline{u}$ (its direction of variation, which excludes the static offsets) over the 19 h contiguous stream of 2026-10-04/05 (2.8 million batches). In that stream A and B lay at 90.50° and 269.50° relative to $\underline{D}$. Rotating by the default constants leaves the in-phase variation unchanged and reduces the quadrature standard deviation of x from $1.8 \cdot 10^{-4}$ to $1.1 \cdot 10^{-4}$ (S_1) and from $4.0 \cdot 10^{-4}$ to $0.9 \cdot 10^{-4}$ (S_2), against an in-phase standard deviation of $1.4 \cdot 10^{-3}$ and $2.5 \cdot 10^{-3}$. The axis of S_2

is stable between 4 h segments ($-9.1°$ to $-10.4°$); that of S_1 is less certain ($-7°$ to $-8°$ in most segments, $-6.0°$ over everything), so $\pm 1.5°$ is the honest uncertainty. These are estimates from drift and noise, not from a controlled tilt: determine δ_1, δ_2 properly from a deliberate tilt step and set them over the API (Calibrations 0x0D, 0x0E); there is no on-instrument routine, and the flip calibration does not touch them.

Consequences. The gain of the reading is no longer reduced by $\cos\delta$ (0.6 % for S_1, 1.3 % for S_2 with the defaults), so a k calibrated before this change is off by that amount (k is to be redone anyway, Section 13.3). The zero level is not affected in principle, but the static quadrature offset of $\underline{u}$ leaks into the in-phase reading differently after the rotation, so the zero calibration has to be repeated as well. The change of the EEPROM layout reseeds the whole displacement page to its defaults: gains, sensitivities, zero offsets and the invert flags have to be entered again.

10 Sampling

All four signals (S_2, B, A, S_1) are digitised by one ADS131M04, a four-channel, 24-bit delta-sigma ADC with simultaneous sampling.

10.1 Clocks and locking to the sine

The sampling rate is locked to the sine because both are derived from the *same* clock, with fixed integer ratios (Figure 6). The microcontroller's 64 MHz system clock (from the 8 MHz crystal via the PLL) feeds two timers. Each divides by 12 and outputs a 5.3333 MHz square wave: TIM1 clocks the DDS (Section 3), TIM2 clocks the ADC.

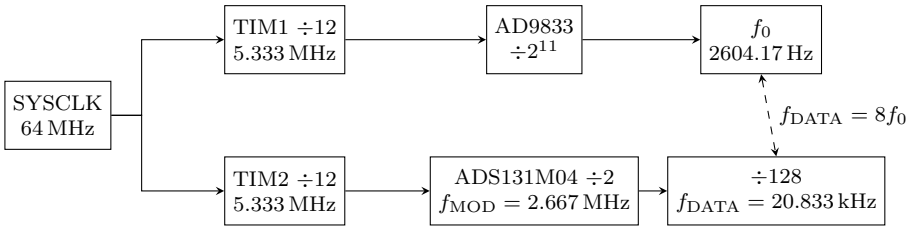


Figure 6: Clock tree. DDS output and ADC sample rate come from the same 64 MHz clock through fixed dividers.

The DDS output frequency is $f_0 = f_{\mathrm{MCLK}}/2^{11}$ (FREQREG $= 2^{17}$). The ADC runs its modulators at $f_{\mathrm{MOD}} = f_{\mathrm{CLKIN}}/2$, and the firmware selects an oversampling ratio of OSR = 128 (register `CLOCK`), so one conversion per channel is delivered at

$$f_{\mathrm{DATA}} = \frac{f_{\mathrm{CLKIN}}}{2\,\mathrm{OSR}} = \frac{f_{\mathrm{CLKIN}}}{256} = 20.833\,\mathrm{kHz} = 8\,f_0. \tag{11}$$

Exactly **eight samples per sine period**, always: a change of the crystal frequency (initial tolerance or temperature drift) moves f_0 and f_{DATA} together. There is no other oscillator and no software timing involved in choosing the sampling instants.

The software only *reads* conversions. A third timer (TIM7) polls the ADC's data-ready line at $2f_{\mathrm{DATA}}$ (also derived from the same clock) and starts a DMA read of the 18-byte SPI frame (status + 4 × 24-bit data + CRC, about 10 µs at 16 MHz). The data-ready line stays active until read, so a late read delays a sample but never changes when it was taken.

What is not locked. The *phase* between the DDS sine and the sample grid is fixed while the instrument runs, but arbitrary: it depends on when each device was started. Only phases of the four channels *relative to each other* are therefore meaningful in general (see Section 10.5).

10.2 Simultaneous sampling of the four channels

The ADS131M04 contains four independent delta-sigma modulators with their own digital filters, all running from the same modulator clock. There is no multiplexer: the four channels are converted in parallel and their results are updated together, at the same falling edge of the data-ready line. One SPI frame therefore contains four samples that belong to the same instant.

ADC channel	Pin	Signal	PGA gain
CH0	AIN0P	S_2 (sensor 2)	16
CH1	AIN1P	B	1
CH2	AIN2P	A	1
CH3	AIN3P	S_1 (sensor 1)	16

All negative inputs are grounded (single-ended use). The chip also has per-channel phase-delay registers; they are left at their default of zero, so no artificial inter-channel delay exists.

10.3 PGA gain and resolution

Each channel has its own programmable-gain amplifier. The available gains and the resulting full-scale range (FSR $= \pm 1.2\,\mathrm{V}/G$) and code size ($1\,\mathrm{LSB} = 2.4\,\mathrm{V}/(G \cdot 2^{24})$) are:

Gain G	FSR (peak)	1 LSB
1	±1200 mV	143 nV
2	±600 mV	71.5 nV
4	±300 mV	35.8 nV
8	±150 mV	17.9 nV
16	±75 mV	8.94 nV
32	±37.5 mV	4.47 nV
64	±18.75 mV	2.24 nV
128	±9.375 mV	1.12 nV

Used: gain 1 for A and B (their ≈ 0.63 V peak is already about half of full scale, so any more gain would clip) and gain 16 for S_1 and S_2 (full scale ±75 mV, corresponding to about ±2.5 mm/m, Section 9). The register value is `0x4004` (`GAIN1`). The PGA gain is an independent stage from the software gain constants used later in the ratio calculation.

Resolution. Samples are 24-bit two's-complement codes, sign-extended to 32 bit in the firmware; full scale is $\pm 2^{23}$ codes. The 24 bits are the code width, not the useful resolution: the datasheet quotes a dynamic range of 102 dB at gain 1 for a data rate of 4 kSPS, and less at our 20.8 kSPS. An earlier bench capture of unconnected inputs at gain 1 showed a noise floor of about 150 codes RMS (≈ 21 µV) per sample; averaging in the digital processing improves on this. The input impedance is $330\,\mathrm{k\Omega} \cdot 4.096\,\mathrm{MHz}/f_{\mathrm{MOD}} \approx 510\,\mathrm{k\Omega}$ at gains 1–4 and higher (at least 1 MΩ) at gains 8 and up.

10.4 Digital filter and aliasing

Each channel is decimated by a third-order sinc filter (the chip's regular data path). For OSR = 128 its properties at the excitation frequency are (standard sinc^3 results, not measured):

- **Gain:** 0.925 (−0.67 dB) — the amplitude droop of the filter at $f_0 = f_{\mathrm{DATA}}/8$.
- **Delay:** $3(\mathrm{OSR}-1)/(2f_{\mathrm{MOD}}) = 71.4$ µs, i.e. a phase lag of 67° at f_0.

Both are identical for all four channels, so they do not change the relative phases or the ratios; they only affect absolute amplitude and absolute phase. Signals near $kf_{\mathrm{DATA}} \pm f_0$ would alias onto f_0; the filter attenuates $f_{\mathrm{DATA}} - f_0 = 18.2$ kHz by 51 dB, on top of the analog low-pass

filters in front of the ADC. (After a chip reset the first two conversions come from a faster, less selective filter path; this only matters at power-up, long before the acquisition is started.)

10.5 Sample integrity and phase reference

The eight-samples-per-cycle structure only works if no conversion is lost or duplicated. Every frame is therefore checked:

- its first word must equal the expected status word (framing),
- its CRC word must match the CRC computed over the frame,
- the DMA ring must not be overrun by the reader,
- the number of frames received must track elapsed time times f_{DATA} (both from the same clock, so the expectation is exact); a deviation of more than 8 frames for 200 ms counts as a lost or duplicated conversion.

Any of these latches a fault, logs an error and stops the acquisition; a measurement never continues silently with a shifted sample grid. (This is different from the input and output drop counters seen in the test logs: the input counter counts carrier cycles lost when a whole 64-cycle batch is dropped because the software batch ring was full, which loses averaging time but keeps the sample grid intact. Since fw 0.10.64 the unit dropped is always a whole batch; before that it was a single 8-sample cycle.)

Phase reference. Within one acquisition run the phase of each channel in the digital frame is stable (standard deviation of the phase of A over an hour: below 0.1°). Across runs it is *not guaranteed* by design: the running sample index that defines the phase reference is reset when the firmware initialises, not each time the acquisition is started, while the ADC keeps converting in between. Two runs on two boots gave nearly the same phase of A in this frame (122.85° and 122.67°), but a third run gave 334.39°: **the absolute phase is not reproducible across runs**, as the design implies, and nothing may depend on it. (The phase of B relative to A was the same in all three, $\approx 179°$.) For the planned phase calibration (Section 9) this means: phases relative to each other are always meaningful, and using the measured difference phasor $\underline{D} = \underline{A} - \underline{B}$ as the run-time reference (rotate every phasor so that $\underline{D}$ is at 90°) removes any dependence on the sample-index alignment.

11 Amplitude and Phase from the Samples

Each channel delivers eight samples per sine period, exactly 45° apart. From them we want the amplitude and phase of the f_0 component. This is the central step of the instrument.

11.1 The formula

Let $x[n]$ be the samples of one channel, $n = 0, 1, 2, \ldots$ counting from the start, and $\theta = 2\pi/8 = 45°$. The signal is

$$x[n] = X_{\mathrm{dc}} + |X| \sin(\theta n + \varphi) + \text{noise}, \tag{12}$$

with unknown amplitude $|X|$ and phase φ (and an unwanted DC term). Multiply each sample by $\cos \theta n$ and $\sin \theta n$ and add up over a whole number N of periods, i.e. over $M = 8N$ samples:

$$I = \sum_{n=0}^{M-1} x[n] \cos \theta n, \qquad Q = \sum_{n=0}^{M-1} x[n] \sin \theta n. \tag{13}$$

For the signal above, using $\sin\alpha\cos\beta$ and $\sin\alpha\sin\beta$ identities and the fact that $\sum\cos 2\theta n = \sum\sin 2\theta n = 0$ over whole periods,

$$I = \tfrac{M}{2}\,|X|\sin\varphi, \qquad Q = \tfrac{M}{2}\,|X|\cos\varphi, \tag{14}$$

and therefore

$$\boxed{|X| = \frac{2}{M}\sqrt{I^2+Q^2}, \qquad \varphi = \operatorname{atan2}(I,\,Q).} \tag{15}$$

This is the closed formula. It needs no iteration and no fitting routine: two weighted sums per channel and, only if the angle itself is wanted, one arctangent. It is the single-bin discrete Fourier transform at f_0, and equivalently the exact least-squares fit of a sine wave plus DC to the samples, because for uniformly spaced samples over whole periods the fit's normal equations are diagonal.

The weights are only these eight pairs:

n	0	1	2	3	4	5	6	7
$\cos\theta n$	1	0.707	0	−0.707	−1	−0.707	0	0.707
$\sin\theta n$	0	0.707	1	0.707	0	−0.707	−1	−0.707

Phase convention. The firmware treats the pair as a complex number $P = I + \mathrm{j}Q$ and reports $\psi = \operatorname{atan2}(Q, I)$. For the signal above $\psi = 90° - \varphi$: the firmware angle *increases when the signal is delayed*. Only differences between channels are used, and these are the delay of one channel behind the other: $\psi_S - \psi_B = -(\varphi_S - \varphi_B)$. Sections 3–9 used the physical convention (positive phase = leads); all measured phases quoted from the firmware are delays, as labelled where they occur.

11.2 How many periods

The formula is exact for *any* whole number N of periods, and a non-integer number of periods gives an error. The firmware always uses whole periods: it sums $N = 64$ consecutive periods (`DISPLACEMENT_BATCH_CYCLES`), i.e. $M = 512$ samples or 24.6 ms per result, about 40 results/s. N trades noise against speed:

N	$M = 8N$	Time	Noise factor $\sqrt{2/M}$
1	8	0.38 ms	0.50
8	64	3.1 ms	0.18
64	512	24.6 ms	0.0625
512	4096	197 ms	0.022

For white noise of RMS σ_x per sample, the amplitude estimate has a standard deviation $\sigma_x\sqrt{2/M}$, and so does $|X|\cdot\sigma_\varphi$ for the phase (in radians). With $N = 64$ this is $0.0625\,\sigma_x$: a noise of 150 codes per sample (an older bench figure at gain 1) becomes about 9 codes. A simulation confirms the formula (9.3 codes found). The averaging window also defines the estimator's bandwidth, $1/T = 40.7$ Hz: anything that changes faster than that is blurred, and tones at $f_0 \pm k\cdot 40.7$ Hz are rejected exactly. Larger N lowers noise but slows the result and averages over any real change of the tilt.

11.3 Does it matter where in the wave we sample?

No. Sampling at $0°, 45°, 90°, \ldots$ or at $10°, 55°, 100°, \ldots$ is equivalent, and no offset is superior. A shift α of the sample grid is the same as changing the signal phase from φ to $\varphi + \alpha$: the same eight equally spaced samples per period are used, the weights remain orthogonal, so the amplitude and the noise of the result are unchanged and only the reported phase moves by α. A

simulation with the firmware's integer weights confirms this: for offsets of 0°, 10°, 22.5°, 45° and 67° the amplitude is identical to six digits, the noise of the estimate is the same (9.1 to 9.5 codes at 150 codes input noise), and the phase shifts by exactly the offset.

What does matter:

- **Equal spacing and a whole number of periods.** Both are guaranteed by the shared clock (Section 10) and by summing whole cycles. This is what makes DC (exactly) and the 2nd to 6th harmonics vanish from the result.
- **Noise well above one LSB.** The same eight sample positions repeat every period. A noiseless signal would produce the same quantisation error every period, which averaging cannot remove, and then the grid offset would matter slightly. Here the noise is far above one code, so the quantisation error is randomised.
- **No clipping** (a clipped peak is no longer a sine).

11.4 Limits

- **Aliased harmonics.** With eight samples per period the 7th and 9th harmonics (18.2 kHz, 23.4 kHz), and the 15th, 17th, ..., fold exactly onto the fundamental and are *not* rejected by the formula (the simulation shows them passing with full weight). They are suppressed only by the analog filters and the ADC's sinc filter (−51 dB and −58 dB at 7th and 9th). The excitation is a DDS sine and the A/B channels are clean (2nd/3rd harmonic −50/−58 dBc), but the *sensor* channels carry large low harmonics in the 24 h raw-ADC capture (S_2: 2nd −21, 3rd −15 dBc; S_1: −37, −32 dBc). The 2nd to 6th are rejected exactly, but the 7th and 9th cannot be seen in an 8-sample capture, so whether they matter for the sensors is **unchecked**; "negligible" is shown only for A and B. A capture with a faster sample clock (or a scope) would settle it.
- **Integer weights.** The firmware uses 14-bit fixed-point weights (16384 and 11585 $\approx 16384/\sqrt{2}$). The table sums are exactly zero, so DC is still rejected exactly. The rounding makes the overall amplitude $1.0 \cdot 10^{-5}$ too small and lets the 3rd harmonic leak at 10^{-5}. The scale error is the same on all channels and cancels in ratios.

11.5 Implementation

Sample level (integer, one add per sample). Every ADC sample of all four channels is added into a plain 32-bit sum for its position n mod 8 in the carrier cycle, over a whole batch of $N = 64$ cycles: $s_n = \sum_{c=0}^{N-1} x[8c + n]$, $n = 0 \ldots 7$, one accumulator per position and channel. There is no multiplication, no sign handling and no 64-bit arithmetic in the sample path. (Until fw 0.10.64 each sample was multiplied by its Q14 weight and added to a 64-bit I/Q sum; on the Cortex-M0+, which has no 64-bit multiply, that was eight library calls per ADC sample and about a fifth of the CPU.)

Why 32 bits are enough. A position sum adds N signed 24-bit codes, $|s_n| \le N \cdot 2^{23}$, which fits a 32-bit signed integer for any $N \le 256$ (checked at compile time). The bound is deterministic, so no further headroom is needed. With the 64-cycle batch the worst case is 2^{29}.

Batch level (integer, exact). When the 64th cycle is complete the 32 sums are queued, and the task applies the weights once per batch. The Q14 weights for positions $n = 0 \ldots 7$ are

$$\cos \theta n \to \{16384,\ 11585,\ 0,\ -11585,\ -16384,\ -11585,\ 0,\ 11585\},$$
$$\sin \theta n \to \{0,\ 11585,\ 16384,\ 11585,\ 0,\ -11585,\ -16384,\ -11585\},$$

so by linearity

$$I = 16384\,(s_0 - s_4) + 11585\,(s_1 - s_3 - s_5 + s_7), \qquad Q = 16384\,(s_2 - s_6) + 11585\,(s_1 + s_3 - s_5 - s_7), \tag{16}$$

evaluated in 64-bit integers (the four-term groups can reach 2^{31} and the products about 2^{45}; there are two real 64-bit multiplications per channel per *batch*). This gives bit-for-bit the same integers as the per-sample weighting it replaced (verified on random and full-scale data), so nothing downstream changes. The result per channel and batch is a pair (I, Q) with magnitude $|I + \mathrm{j}Q| = 65536 \cdot N \cdot X_{\mathrm{code}}$, where X_{code} is the peak amplitude in ADC codes ($65536 = 4 \cdot 16384$ is the weight scale times the factor $M/2 = 4$ per period); converting to volts uses the LSB of the channel's PGA setting (Section 10.3). The ratio stage that follows uses the complex numbers (I, Q) directly and does not need the angle or the amplitude separately.

Dropped data. The queue holds 4 batches (about 98 ms). If the task falls further behind, the newest *whole batch* is dropped and counted (64 cycles); there are no partial batches. The per-cycle sequence counter advances regardless, so a drop shows up downstream as a jump of a whole number of batches, and the sample grid is never shifted.

12 From Phasors to Tilt: the Complex Ratio

12.1 Two ways to get the tilt

Let $\underline{A}$, $\underline{B}$, $\underline{S}$ be the measured phasors (Section 11). The obvious method is:

1. decide the sign from the phase of $\underline{S}$ (closer to A or to B), and
2. take $|\underline{S}|$ and divide by the sensitivity (20 µV RMS per µm/m).

Firmware status. Up to fw 0.10.61 the firmware evaluated the full complex ratio below, including the $-\underline{B}$ term. From fw 0.10.62 it evaluates the projection $\mathrm{Re}[\underline{S}/(k(\underline{A} - \underline{B}))]$ (the $\lambda = 0$ estimator of Section 12.7), because the $-\underline{B}$ term injected the A/B balance drift into the reading. Sections 12.1 to 12.6 describe and compare the original ratio, which explains why the change was made. The original firmware evaluated a complex division per sensor:

$$\underline{x} = \frac{\underline{S}/k - \underline{B}}{\underline{A} - \underline{B}}, \qquad \text{tilt} \propto \mathrm{Re}\,\underline{x} - \tfrac{1}{2}, \qquad \text{residual} = \mathrm{Im}\,\underline{x}, \tag{17}$$

with k a constant (one software setting per sensor, since fw 0.10.67 no longer split into attenuator and gain) that brings $\underline{S}$ to the scale of $\underline{A}$ and $\underline{B}$. The output is delta $= 2\,d_0\,(\mathrm{Re}\,\underline{x} - \frac{1}{2}) - \text{zero}$, where d_0 (built from the nominal sensitivity and a per-sensor calibration factor) and the zero offset are calibration constants.

Meaning of x. The sensor is a capacitive divider between the two electrodes. In the ideal case its output is $\underline{B} + x(\underline{A} - \underline{B})$ with $x \in [0, 1]$ the position of the pendulum plate between them: $x = 0$ at B, $x = 1$ at A, $x = \frac{1}{2}$ at zero tilt. Eq. (17) is the inverse of that model: it recovers x from the measured phasors. Because the plate position changes by only a tiny fraction for each µm/m, $x - \frac{1}{2}$ is of the order of 10^{-6} per µm/m.

12.2 Advantages and disadvantages

Complex ratio with the B term (fw $\leq$ 0.10.61)	Magnitude + sign (naive)
Advantages of the complex ratio	
Ratiometric: DDS amplitude (14 % above the sensor's nominal supply, 200 ppm/°C drift), filter and op-amp gain and the ADC reference cancel between $\underline{S}$ and the references.	Needs the absolute excitation amplitude: sensitivity is quoted at 2 V RMS, so any excitation change is a scale error.
No absolute phase needed: sample-grid alignment and the filter delay drop out of the division.	Needs a phase reference to decide the sign, i.e. a calibrated absolute phase.
Linear through zero and unbiased: noise averages to zero.	Nonlinear at zero: $\lvert\underline{S}\rvert$ of a noisy zero signal is positive (Rayleigh bias) and the sign flips randomly. The instrument works around zero, so this matters most.
Residual $\operatorname{Im}\underline{x}$ is a built-in consistency check (phase errors, interference, noise).	No consistency check.
Phase errors of $\underline{S}$ are second order in the reading ($\cos\delta$).	$\lvert\underline{S}\rvert$ is completely insensitive to phase errors.
Compensates an imbalance of $\underline{A}$ and $\underline{B}$ at the sensor: the sensor passes $(\underline{A}+\underline{B})/2$ into $\underline{S}$ and the formula subtracts it (needs the right k, see below).	Cannot compensate it; an imbalance shifts $\lvert\underline{S}\rvert$ directly.
Disadvantages of the complex ratio	
The compensation is only as good as k (magnitude *and* phase of the gain ratio S path : A/B path). A wrong k under-compensates, and a mismatch between the A and B *measurement* paths is not compensated at all (next sections).	Zero does not depend on how $\underline{A}$ and $\underline{B}$ are measured.
Needs accurate phase alignment between the paths.	Needs the absolute gain and the exact sensitivity.
More arithmetic (complex, floating point on a Cortex-M0+).	Trivial.

A middle way exists: project $\underline{S}$ on the reference, $\operatorname{Re}[\underline{S}/(k(\underline{A}-\underline{B}))]$. It is ratiometric, phase-insensitive to first order, linear and unbiased like the full ratio, but it drops the $\underline{A}+\underline{B}$ term: it gives up the compensation of a real imbalance and in return becomes immune to a mismatch of the two measurement paths. Which is better depends on which effect dominates. The data (Sections 12.3 to 12.6) favour the projection, and it has been the firmware's estimator since fw 0.10.62. On a 74-minute contiguous phasor stream the two estimators give for S_1/S_2 a detrended standard deviation of 1.64/1.70 µm (full ratio) against 0.56/0.57 µm (projection), and an Allan deviation at 60 s of 0.90 $\rightarrow$ 0.30 µm for S_1.

12.3 Imbalance of A and B: what is and is not corrected

Writing $\underline{x}$ out,

$$\underline{x} - \tfrac{1}{2} = \frac{\underline{S}}{k(\underline{A}-\underline{B})} \; - \; \frac{\underline{A}+\underline{B}}{2(\underline{A}-\underline{B})}. \tag{18}$$

The second term, $\underline{z}$, is zero when $\underline{A}$ and $\underline{B}$ are exactly antiphase and equal. Its purpose is to remove the effect of an imbalance on the sensor: the sensor is a divider between its two supply voltages, so its output contains the midpoint $(\underline{A}+\underline{B})/2$ plus the tilt signal $(x-\frac{1}{2})(\underline{A}-\underline{B})$. If the excitation at the sensor terminals becomes imbalanced, $\underline{S}$ changes, and because $\underline{A}$ and $\underline{B}$ are measured, the software can and does subtract that change. **This works automatically, but only if k is right:** the measured $\underline{S}/k$ must equal the same midpoint in the scale of the measured $\underline{A}$, $\underline{B}$, i.e. k must be the gain ratio S path : A/B path, in magnitude and in phase. (Assumed here: the sensor passes the full midpoint into its output; parasitic capacitance would

reduce this and lower the required k by the same factor.) If k is a factor f too large, only the fraction $1/f$ of the imbalance is cancelled by $\underline{S}$ and the rest appears in the reading.

Two things are *not* corrected in any case. An imbalance of the *measurement* paths (attenuator resistors, coupling capacitors, ADC channels 1 and 2) changes the measured $\underline{A} + \underline{B}$ but not the sensor's $\underline{S}$, so $\underline{z}$ moves and nothing compensates it. And since $\underline{S}_1$ and $\underline{S}_2$ share the same $\underline{A}$ and $\underline{B}$, all these effects are *common to both sensors* and cancel in the difference $S_1 - S_2$.

The logged data show that the slow common-mode drift of the sensor readings follows this term. From the phasors in the logs, $\mathrm{Re}\,\underline{z}$ was computed and compared with the minute-averaged raw sensor readings:

	Variation of $\mathrm{Re}\,\underline{z}$	Correlation	Scatter: raw → z removed
24 h granite run (gain 16)	111 ppm	0.86 (S_1), 0.99 (S_2)	26 → 13 µm, 27 → 4 µm
1 h run (gain 1 on all)	10 ppm	−0.99, −0.95	6.0 → 0.6 µm, 5.8 → 1.8 µm

(The removal is a one-parameter regression per sensor on the logged minute-means: it shows how much of the variation follows this term, not a validated correction.) The result is independent of the PGA setting; the differential $S_1 - S_2$ is correspondingly quiet.

There are two explanations. (a) The excitation at the sensor really becomes imbalanced (for example through the resistor ratio R_{56}/R_{55} of the inverter) and the formula under-compensates it because k is too large (next section). (b) The measurement paths of A and B drift against each other (or the sensor output does not carry the midpoint at all), which no choice of k can compensate.

The logged phasors allow a first check, because under (a) the quantity $\underline{u} = \underline{S}/(\underline{A} - \underline{B})$ must follow $\underline{z}$ with slope $-k_{\mathrm{phys}}$ (about −68 at gain 16), and under (b) with slope 0. A regression over the whole 24 h run is inconclusive: the slopes scatter (S_1: −97, S_2: −19) and explain only 10–40 % of the variance, because the real tilt also drifts. A cleaner view is the step at the start of charging ($t \approx 247$ min): within minutes $\mathrm{Re}\,\underline{z}$ jumped by +42 ppm and both readings by about +60 µm, yet $\underline{u}$ of both sensors changed by only $+2.8 \cdot 10^{-4}$ and $+3.5 \cdot 10^{-4}$, whereas (a) with $k = 68$ would have required $-2.9 \cdot 10^{-3}$. So almost 99 % of the jump in the reading is the $\underline{z}$ term itself and $\underline{S}$ did not follow, which favours (b). This is a single event, and a real tilt change cannot be excluded, but both sensors moved almost identically.

To settle it, a controlled test is needed: either repeat forced-charge steps on an undisturbed plate (several clean $\underline{z}$ steps), or, more directly, unbalance the excitation by a known amount (for example a resistor in parallel with R56 changes $|B|$ by about 1 %, a step of some 5000 ppm in $\underline{z}$, far above the drift). The slope of $\underline{u}$ against $\underline{z}$ in that test measures the response of $\underline{S}$ to an imbalance directly, i.e. the complex k (including the sensor's own phase); a slope near zero would mean the sensor output does not carry the midpoint. Grounding the sensor input does not help: the reading is then $\underline{z}$, which the logs already contain.

12.4 Effect of charging on the balance

In the 24 h run the battery charger started at $t = 246.9$ min (USB was connected throughout). Within the same minute the board temperature began to rise (26.5 °C → 30.5 °C in 10 min) and the balance changed:

Phase	Time	Re $\underline{z}$ change	Reading change
before charging (warm-up)	0–247 min	+11 ppm	+12 µm
charging starts	247–256 min	+48 ppm	+55–60 µm
relaxation, temperature steady	258–307 min	−41 ppm	−45–55 µm
second rise, charge current tapering	317–362 min	+23 ppm	+25–30 µm
slow decline	362–1207 min	−48 ppm	−55–65 µm
unexplained heating event	1240–1277 min	−44 ppm	−50–60 µm

One ppm of Re $\underline{z}$ corresponds to about 1.1–1.3 µm in S_1 and S_2. The step at the start of charging changes $\underline{z}$ by +48 ppm (real) and −96 ppm (imaginary), which is a change of $|B|/|A|$ by about 190 ppm and of the phase between them by 0.022°, both within 10 minutes. This size is plausible for resistor temperature-coefficient mismatch: 190 ppm over 4 K is 47 ppm/K. The A and B input networks sit about 30 mm apart on the board and at different distances from the charger (about 58 mm and 69 mm), so a temperature gradient across them is likely.

Charging therefore *did* change the balance, and the change is the largest single feature of the run: the maximum of Re $\underline{z}$ in the 24 h run (111 ppm peak-to-peak) is reached at the end of the charging step. The whole run before charging shows only about 11 ppm. It does not explain everything: the minimum comes from a later heating event without any change in the charging state, with the opposite sign (−44 ppm at +3.4 K), and the slow 13 hour decline continues afterwards with the temperature back at its starting value. So heating events move the balance, but not with a fixed sign or a fixed coefficient per kelvin. All of this is common to S_1 and S_2: the differential $S_1 - S_2$ does not show the ±60 µm steps.

12.5 The scale constant k

k is the gain ratio of the S path to the A/B path in the ADC domain. From the networks and PGA settings (Sections 7, 9, 10),

$$|k_{\text{phys}}| = \frac{16 \cdot 0.903}{1 \cdot 0.212} \approx 68, \tag{19}$$

and, because the two paths delay the signal differently (S path 2.6°, A/B path 23.7° plus the sensor's own delay, Section 9), k is in fact complex, $|k|\, e^{\mathrm{j}\delta}$ with δ of order 10° to 20°. Up to fw 0.10.66 the firmware used a real k = attenuator 3.000× gain 160 = 480: a nominal attenuator of 3 (in reality the A/B network attenuates by $1/0.212 = 4.7$ at f_0) and an assumed external gain of 10 (there is none; the S network has a gain of 0.9) times the PGA of 16. The factor between that and the network-derived magnitude is 7.0, the same as the empirical sensitivity correction found earlier with a paper-shim test. Since fw 0.10.67 $|k|$ is simply one empirical number per sensor with no further interpretation; since fw 0.10.68 it is stated and stored for a PGA gain of 1 (the firmware divides $\underline{S}$ by the PGA first); and since fw 0.10.69 it is the sensitivity k of Section 13.3 (default 0.0213 per mm/m, derived there from the nominal sensor sensitivity), the separate d_0 and sensitivity constants having been removed.

This is harmless for the tilt term, because d_0 compensates it, but it matters for the imbalance:

- **Real imbalance:** with $k = 480$ instead of 68 only about $68/480 = 14$ % of it is cancelled; 86 % appears in the reading. With the correct k it would be cancelled.
- **Measurement-path mismatch:** the weight of $\underline{z}$ is $2d_0 \propto k$ for a fixed tilt scale, so a too-large k amplifies its effect by the same factor 7; with the correct k it shrinks 7× but does not vanish.
- **Phase:** a real k leaves the imaginary part of the compensation uncancelled (visible in the residual) and a fraction $1 - \cos\delta$ of the real part (1.5 % at 10°). The planned phase

calibration (Section 9) removes this by rotating $\underline{S}$ onto the real axis, which is the same as making k complex.

In both cases, therefore, the network-derived k is the better choice, and it is a software setting only (the gain constants can be set over the API).

12.6 Correcting the imbalance: results on the 24-hour data

The 24 h data contain what is needed to remove the imbalance term. The reading contains $c \cdot \mathrm{Re}\,\underline{z}$, where c was taken from the step at the start of charging ($\mathrm{Re}\,\underline{z}$ +41.9 ppm, both readings +59.5 µm, so $c = 1.42$ µm/ppm for both sensors). This step is faster than any real tilt change, so c does not depend on the slow drift it is applied to. The corrected reading is reading $- c\,\mathrm{Re}\,\underline{z}$. Fitting c over the run instead gives similar values (1.4–1.7 µm/ppm) but also absorbs real tilt changes that happen to correlate with z.

The last hours of the run are excluded: the bump around 21 h is very likely direct sunlight on the sensors, a genuine sensor input that must not be corrected away (it also heated the board and moved z); its onset appears to start around 17.5 h, where S_1 and $S_1 - S_2$ turn upward. Drift of the 1-minute means, relative to the first 10 minutes:

Window	Series	p–p (µm)	std (µm)	trend (µm/h)	detrended rms (µm)
0–20 h	S_1 raw	118	25.2	−3.58	14.5
	S_1 corrected	35	9.7	−1.57	3.6
	S_2 raw	101	19.6	−2.00	15.8
	S_2 corrected	12.4	2.5	+0.02	2.5
	$S_1 - S_2$ (unchanged)	38.3	9.5	−1.58	2.7
0–17.5 h	S_1 raw	111	22.8	−3.35	15.2
	S_1 corrected	35	9.9	−1.94	1.5
	S_2 raw	92	18.3	−1.58	16.5
	S_2 corrected	10.8	2.2	−0.17	2.1
	$S_1 - S_2$ (unchanged)	38.3	9.2	−1.77	1.9

Cross-validation (coefficient fitted on one half of the 20 h and applied to the other) improves both sensors in both directions (S_1: 12.8 → 4.4 and 14.3 → 7.2 µm; S_2: 9.5 → 3.0 and 17.7 → 4.4 µm std). What the correction does and does not do:

- S_2 becomes flat: the std drops by 8×, the linear trend vanishes. S_1 loses most of its scatter but keeps a smooth trend of about −1.6 to −1.9 µm/h.
- $S_1 - S_2$ does not change: z is common to both sensors and c is equal. Its remaining slow drift is the difference of the sensors' own drift, not an electronic imbalance effect.

Local noise. The correction also removes short-term noise, because the wander of z on time scales from seconds to minutes appears in both readings. Allan deviation of the readings (quiet segments, charging transient and sunlight excluded), in µm:

Averaging time	S_1 raw	S_1 corr.	S_2 raw	S_2 corr.	$S_1 - S_2$
1 s	1.29	0.96	2.74	2.60	1.67
10 s	0.99	0.35	1.30	0.90	0.57
60 s (1 minute)	**0.95**	**0.27**	**1.02**	**0.45**	0.25
5 min	1.04	0.27	1.03	0.30	0.19
30 min	2.52	0.84	2.34	0.59	0.93

Over 1-minute windows the minute-to-minute jitter of S_1 falls from 0.95 to 0.27 µm (3.5×) and that of S_2 from 1.02 to 0.45 µm (2.3×); at 5 minutes the improvement is 3.9× and 3.4×. The scatter of the individual 24.6 ms readings inside a 1-minute window changes much less

(median S_1 2.65→1.55, S_2 4.53→4.06 µm): that is sensor and ADC noise, not imbalance. The differential $S_1 - S_2$ (unchanged) reaches 0.25 µm at 1 min, which is about the level the corrected individual sensors approach. (Scripts: `Testing/2026-09-27_24hr_granite_plate_test/correct_drift_with_ab_balance.py` and `noise_quantification.py`.)

12.7 Proposed change to the estimator

The current formula gives the correction term $\underline{z}$ the full weight $\lambda = 1$, which is right only if the sensor output carries the midpoint $(\underline{A} + \underline{B})/2$ at the scale k. At the charging-start step, $\underline{S}/(\underline{A} - \underline{B})$ changed by at most about 10 % of what full compensation would require (and with the wrong sign), so $\underline{S}$ carries little or none of the imbalance: the term $\underline{z}$ mostly injects the measured imbalance into the reading. (Confirmed so far by a single natural event; a deliberate imbalance would settle it.) A generalised estimator gives $\underline{z}$ a calibrated weight,

$$\text{tilt} \;\propto\; \mathrm{Re}\Big[\frac{\underline{S}}{k(\underline{A} - \underline{B})} + \lambda\,\underline{z}\Big] - \text{zero}, \tag{20}$$

with $\lambda = 1$ the current estimator, $\lambda = 0$ the projection estimator ($\underline{S}$ divided by $\underline{A} - \underline{B}$ only), and intermediate or complex values for a sensor that passes part of the imbalance. The data favour $\lambda \approx 0$. Proposed steps, in order:

1. **Set** $\lambda = 0$: drop the subtraction of $\underline{B}$ in the ratio. The tilt term is unchanged, so d_0 and the sensitivity constants stay; the zero offset must be calibrated again because it absorbs the former $\frac{1}{2} - \underline{z}$ baseline.
2. **Make** k **complex** (the planned phase calibration), so that the in-phase projection is exact and the quadrature part becomes a clean diagnostic.
3. **Verify** λ with a deliberate, known imbalance, and set it to the measured value if a sensor turns out to pass part of it.

Not changed by this: the ratiometric cancellation of the excitation (division by $\underline{A} - \underline{B}$), and the sensors' own slow drift visible in $S_1 - S_2$. Path gain differences between the S path and the A/B path (PGA, networks) are not cancelled either; the common wander of $|\underline{A}|$ and $|\underline{B}|$ (about 1000 ppm) reaches the reading as a scale error, which is negligible at small tilts. Estimating λ online (for example by regression on $\underline{z}$ in a sliding window) is not recommended, because real tilt correlates with z as well; a fixed, calibrated λ is safer. **Status: step 1 ($\lambda = 0$) implemented in fw 0.10.62; the zero calibration has to be redone (not yet done); steps 2 and 3 open.** One new observation: $|\underline{D}| = |\underline{A} - \underline{B}|$ itself drifted by about 160 ppm in the 74-minute stream (906 ppm over 17.5 h); the projection is ratiometric against it, so this cancels if it is excitation drift, but not if it is drift of the A/B measurement path. A constant-tilt test is needed to tell which.

12.8 What "drop the B term" and "fix k" mean

Status: the B term has been dropped in fw 0.10.62; this subsection records what was changed. The "fix k" parts are still open. Up to fw 0.10.61 the firmware evaluated, per sensor and batch,

$$\underline{x} = \frac{\underline{S}/k - \underline{B}}{\underline{A} - \underline{B}}, \qquad \text{tilt} = 2\,d_0\big(\mathrm{Re}\,\underline{x} - \tfrac{1}{2}\big) - \text{zero}, \qquad k = 3{\times}160 = 480 \text{ (attenuator times gain, until fw 0.10.66).} \tag{21}$$

Expanding, $\underline{x} - \frac{1}{2} = \underline{S}/(k(\underline{A} - \underline{B})) + \underline{z}$ with $\underline{z} = -(\underline{A} + \underline{B})/(2(\underline{A} - \underline{B}))$ (Section 12.3). The two proposed steps act on different parts of this.

Drop the B term.

- *What it is:* in the numerator, use $\underline{S}/k$ instead of $\underline{S}/k - \underline{B}$, i.e. $\underline{x}' = \underline{S}/(k(\underline{A} - \underline{B}))$ and tilt $= 2\, d_0 \,\mathrm{Re}\, \underline{x}'$ −zero. In `compute_sensor_delta()` it means removing the two subtractions of the B phasor from the numerator; the shared reciprocal of $\underline{A} - \underline{B}$ stays. $\underline{B}$ then enters only through the difference $\underline{A} - \underline{B}$.
- *What it does:* it removes the term $\underline{z}$ from the reading, so the reading no longer depends on the balance $(\underline{A} + \underline{B})$ of the two excitation measurements. That term is what the correction in Section 12.6 removed offline: the drift of S_2 dropped by 8× and of S_1 by 4×, and the 1-minute jitter by 2 to 4×. The division by $\underline{A} - \underline{B}$ remains, so the reading stays ratiometric in excitation amplitude and phase.
- *What stays the same:* the tilt term is identical, so d_0 and the sensitivity constants remain valid.
- *What changes:* the stored zero offsets. The static value of $\mathrm{Re}\, \underline{z}$ is about 2500 ppm, which at the present calibration ($c = 1.42$ µm/ppm) is an offset of roughly 3.6 mm/m inside the current zero calibration. After the change the zero calibration must be repeated (the 180° reversal), otherwise the readings are off by about that amount.
- *What it gives up:* compensation of a *real* imbalance of the excitation at the sensor, if the sensor passed it through. The data show that it does not (Section 12.3). The generalised form with a calibrated weight λ (Section 12.7) contains both cases: $\lambda = 1$ today, $\lambda = 0$ here.

Fix k. k is the gain ratio of the S path to the A/B path in the ADC domain, needed so that $\underline{S}/k$ is on the same scale as $\underline{A}$ and $\underline{B}$. It has a magnitude and a phase.

- *Magnitude.* The firmware value 480 was built from a nominal attenuator of 3 and an assumed external gain of 10 times the PGA of 16 (since fw 0.10.67 simply stored as one number $|k|$ per sensor, see Section 13.3). The networks give about 68 on REV B (and about 47 with the planned REV C network, Section 14). The effect depends on whether the B term is kept: with the B term, k decides how much of a real imbalance is cancelled (14 % today, Section 12.5); *without* it, k only multiplies with d_0 in tilt $= (2d_0/k)\,\mathrm{Re}[\underline{S}/(\underline{A} - \underline{B})]$, so only the ratio d_0/k matters, which the sensitivity calibration sets anyway. Fixing the magnitude of k is then cosmetic: the reading does not change, but the internal quantity $\underline{x}'$ becomes the true fraction and the diagnostics are consistent.
- *Phase.* This is the part that changes the arithmetic in either case: k becomes complex, $k = |k|e^{\mathrm{j}\delta}$ (one constant per sensor, Section 9), i.e. $\underline{S}$ is rotated by $-\delta$ before the division. The effect is small: a scale of $\cos\delta$ (1.7 to 3.1 %, absorbed by the sensitivity calibration) and a leakage of quadrature disturbances by $\sin\delta$, which is negligible here because the quadrature part is 6 to 7× quieter than the in-phase part. What it buys is a clean diagnostic and protection against scale drift if δ changes.

Step	Change in the firmware	Effect on the reading	Recalibration
Drop the B term	remove $-\underline{B}$ in the numerator	removes the $\underline{A}/\underline{B}$ balance drift (4 to 8×) and noise	zero calibration
Complex k (phase)	rotate $\underline{S}$ by $-\delta_{1,2}$	small scale change, cleaner diagnostics	phase constants; sensitivity
k magnitude	set $\lvert k\rvert$ to k_{phys} (not needed: $\lvert k\rvert$ is empirical)	none (with the B term dropped), absorbed by d_0	none

Order of priority: dropping the B term (large effect), then the phase constants (small effect), then the numerical value of k (cosmetic). **Status: B term dropped (fw 0.10.62, zero calibration**

to be redone); complex k (approach 2 of Section 9) implemented in fw 0.10.66; the value of $|k|$ not changed.

13 Processing After the Ratio

This section describes what the firmware does with the phasors after Section 11, in the order of the data flow. The acquisition is off at boot and is started over the API (`Commands`, resource `0x01`).

13.1 Structure and rates

Stage	Unit	Rate	Notes
ADC sample (all four channels)	1 sample	20.83 kHz	Section 10
Cycle	8 samples	2.604 kHz	summed per position (32-bit)
Batch	64 cycles	40.7 Hz	24.6 ms; weights, ratio, delta
Display value	25 batches (Hann)	4.07 Hz	LIVE screen, Sec. 13.5

The completed batches (32 position sums each) are queued in a small ring buffer of four entries. If the consumer falls behind, a whole batch (64 cycles) is dropped and counted; this lowers the averaging but never shifts the sample grid (Section 10.5). The expensive part (the complex division) runs only once per batch; the 64-cycle sum before the division costs only integer additions and gives the same noise as many small batches averaged afterwards, because the ratio is effectively linear at this operating point (x stays within 10^{-5} of 0.5).

13.2 Plausibility checks

- **Clip:** per sample and channel, a code within 1 % of the ADC's digital limit is counted. Clipping is the direct saturation detector.
- **Amplitude fault:** per batch, the magnitude of a phasor must not exceed what a full-scale undistorted sine could produce. This does *not* detect clipping (clipping only lowers the magnitude); it catches gain mismatches, corrupted data or accumulation bugs.
- **Degenerate:** per batch, $\underline{A} - \underline{B}$ exactly zero (the division would be undefined). Not expected in practice.
- **ADC integrity** (Section 10.5) is separate and does stop the acquisition. The three checks above are counters and a one-time log entry; they do not stop the pipeline.

13.3 From ratio to displacement

Calibration model (fw 0.10.69). Per sensor the batch gives the complex number $\underline{u} = \underline{S}/(\mathrm{PGA} \cdot \underline{D})$ with $\underline{D} = \underline{A} - \underline{B}$, rotated by the phase calibration (Section 9.1) to $\underline{x} = \underline{u}\, e^{-\mathrm{j}\delta}$. The in-phase part $r = \mathrm{Re}\,\underline{x}$ is a dimensionless ratio; the quadrature part is the residual. The tilt is

$$\text{tilt [mm/m]} = \frac{r - \text{zero}}{k}, \qquad \text{residual} = \mathrm{Im}\,\underline{x}. \tag{22}$$

There are exactly two constants per sensor besides the phase and the sign:

Constant	Default	Meaning
k (sensitivity)	0.0213 per mm/m, stored as 21300 ($\times 10^{-6}$, integer)	Empirical sensitivity: the in-phase ratio r per mm/m, *stated for a PGA gain of 1.* The firmware divides $\underline{S}$ by the PGA of the channel (16 now, defined once in `drv_ads131m04.h`), so k does not change when the PGA is changed. On-board attenuation, filtering and the instrument's gain are all folded in, with no further interpretation.
zero	0 ppm	Value of r at level, in ppm of the ratio (integer). It lives in the r domain, independent of k: recalibrating k does not move the level point, so a flip calibration (Section 13.7) stays valid.
sign (invert)	off	Per sensor, applied to the final value.

The older constants (a shared attenuator, a per-sensor gain, a theoretical d_0 and a "sensitivity" in µV/(µm/m), with the zero stored in a domain that scaled with them) are gone: they all multiplied the same single number. The differential $S_1 - S_2$ is formed after the sign flips.

Default value of k from the nominal sensitivity. The default is not a measurement but follows from the Wyler figure and the network gains. The sensor output is $\underline{V}_S = \kappa\tau\underline{V}_A$ with $\kappa = 20\,\mu\text{V}/2\,\text{V} = 10^{-5}$ per µm/m (Section 8); since $\underline{B} = -\underline{A}$, $\underline{D} = 2\underline{A}$ and at the sensor $S/D = \kappa\tau/2 = 0.5 \cdot 10^{-5}$ per µm/m, i.e. 0.005 per mm/m. In the ADC the sensor path has the network gain 0.903 (Section 9) and the A/B path 0.212 (Section 7), so

$$k = \frac{0.903}{0.212} \cdot 0.005 \text{ per mm/m} = 4.26 \cdot 0.005 = 0.0213 \quad \text{(PGA 1)}. \tag{23}$$

At PGA 16, as now used, the ratio in the ADC codes is 16× this, 0.341 per mm/m: $|\underline{S}| = 0.68\,|\underline{A}|$ at 1 mm/m. The planned REV C networks (Section 14) give $k \approx 0.0147$ at PGA 1.

Comparison with the first calibrations. The two instruments were calibrated against a known tilt on 2026-09-29, using the older constants. Converted to the model above, those calibrations correspond to $k \approx 0.013$ (S_1) and 0.013 (S_2) at PGA 1, i.e. about 60 % of the nominal 0.0213: the sensors deliver about 12 µV per µm/m instead of the nominal 20, if the network gains are right. Candidate causes (not verified): loading of the exciters by the sensor, a different definition of the quoted supply (per exciter or differential), the midpoint leak of the sensor, and sensor tolerance. The earlier "7.5 and 8.8 instead of 20" overstated the deficit, because it was combined with a wrong scale of 480 (Section 12.5). Calibrate k against a known tilt; the default is only the starting point.

13.4 Retired: per-batch quality flag

Until fw 0.10.71 each batch was judged by the step of the residual $\text{Im}\,\underline{x}$ relative to the previous batch against an exponentially weighted baseline (32 batches), and flagged bad if the step exceeded 3× the baseline. This flag was *removed in fw 0.10.72*; the API's quality bytes (Topic 0x03) now report the window-level verdict of Section 13.5 ("the newest display window was not doubtful").

The reason is the finding from the 24 h raw-ADC analysis: the large steps are not outliers but the sensor's 20 Hz pendulum resonance (next subsection), and the flagged batches are those with a large pendulum amplitude. Discarding them made a test estimate 3 to 10× *worse*, because the pendulum is sampled at the batch Nyquist frequency and alternates sign from batch to batch; the plain average cancels it, and removing the large batches unbalances the cancellation. Flagging is useful as a *vibration indicator* at window level; dropping flagged batches from an average is not recommended.

13.5 Pendulum resonance and the smoothing filter

The sensor channels contain a strong, nearly coherent resonance at about 20 Hz (the pendulum; band 18 to 21.5 Hz, $Q \gtrsim 4$; it is coherent between the two sensors, with S_2 about 3× larger than S_1 in the nominal units, and larger by day). This attribution is consistent with the data but not proven. It explains most of the variance of the readings at 1 to 30 s. The batch rate of 40.7 Hz has its Nyquist frequency at 20.35 Hz, right at the pendulum, and the 64-cycle boxcar passes 19.9 Hz with only −3.7 dB, so the pendulum arrives nearly unattenuated, folded to about the Nyquist frequency. Any interval-based sampling of the live value (for example a topic snapshot at ≤ 20 Hz) aliases it further to low frequencies; the contiguous phasor stream (Section 13.10) avoids this.

The 8-batch boxcar of the next subsection cancels the Nyquist-folded part but has poor sidelobes elsewhere. The raw-ADC analysis and the 19 h contiguous phasor stream (`Testing/2026-09-30_bulk_adc_30s_interval/findings.md`) found that a smooth window over contiguous batches is the right tool: it needs no knowledge of the exact resonance frequency, and dropping individual batches does not work (see the caveat in the previous subsection). Choosing an integer number of pendulum periods does not help because the resonance is broad.

Display stream (fw 0.10.71). The LIVE screen shows a Hann window over the newest 25 contiguous batches (0.61 s, delay about 0.3 s, −3 dB at 1.1 Hz), published every 10th batch (40.7 Hz/10 = 4.07 Hz). The worst-case gain in the stopband is −42 dB at 5 to 10 Hz, −60 dB at 10 to 15 Hz and −73 dB at 15 to 20 Hz, so the resonance is suppressed wherever it sits (18 to 21.5 Hz observed) and so is any other floor or plate mode in that range. The first version (fw 0.10.65) used two cascaded 10-batch boxcars, a triangular window over 19 batches with exact nulls at multiples of 4.07 Hz; on the contiguous stream the 4 Hz scatter was 2.6× (night) to 5× (day) lower than with a plain average of 10, and the Hann window over 25 batches a further 13 to 30 % lower, for 75 ms more delay. A Hann window over 19 batches was within ±10 % of the triangular one.

Doubtful readings. The reading is always shown. A window is flagged "doubtful", and the LIVE screen appends an exclamation mark to the line, when the mean squared batch-to-batch step of $\operatorname{Im} \underline{x}$ over the 25 batches exceeds $K = 6$ times the instrument's quiet floor. The floor is the lowest such value of the last 81 batches seen so far, allowed to creep up so that it would double only in about 20 minutes if every window were noisier (a persistent disturbance cannot raise its own limit). If a sensor stays flagged for 10 minutes, though, the quiet level has really changed (a frozen or dead channel came back, a sensor was swapped) and the floor is re-seeded from the current window instead of staying stuck near zero for hours. On the 19 h stream this flags 0.8 % of the display readings at night and 8 % by day; the flagged readings move 1.8× more to the next output than the others. Nothing is flagged until the floor is trustworthy, i.e. rests on two independent 2 s windows (about 4 s after a start; a first window is never judged against itself), and a dropped batch empties the window, so no window ever spans a gap. Each sensor is flagged separately; the differential is flagged if either is. **Status: implemented in fw 0.10.71, build-verified and replayed on the 19 h data on the host, not yet bench-tested.** Since fw 0.10.72 the API's `delta` values are this display stream (the raw batch value until the first window exists); the 8-batch moving average was removed.

13.6 Retired: moving average

Until fw 0.10.71 the API/display value was a boxcar average of the last 8 batch results (SNR gain $\sqrt{8}$, delay about 98 ms). It was removed in fw 0.10.72: the Hann-25 display stream of Section 13.5 replaced it (better suppression of the 20 Hz resonance and out-of-band noise), and the raw per-batch value stays available on Topic 0x03.

13.7 Zero calibration (180-degree reversal)

The instrument's zero error is separated from the surface tilt φ by reversing it in place:

$$\text{step 1} = \varphi + \text{zero error}, \qquad \text{step 2} = -\varphi + \text{zero error} \;\Rightarrow\; \text{zero error} = \tfrac{1}{2}(\text{step 1} + \text{step 2}). \tag{24}$$

Each step averages 64 batches (about 1.6 s), the surface must be stable, and the result is stored per sensor in EEPROM. This cancels any *static* offsets, including (in firmware before 0.10.62) the static part of the $\underline{A}/\underline{B}$ imbalance term $\underline{z}$. The stored zero offsets from before fw 0.10.62 contain about 3.6 mm/m of that term and are **no longer valid**; the zero calibration has to be repeated. The first cycle after the start of the acquisition can be corrupt (stale conversion) and should not enter a calibration or a precision measurement (see Section 15).

Per sensor or both (fw 0.10.70). A flip calibration covers the sensors selected for the run: S_1 only, S_2 only, or both. The sensor not selected keeps its stored zero (both are still averaged during the run, but only the selected zeros are written). Over the API the step-1 command (Commands 0x06, action 1) takes an optional second payload byte, the sensor mask (bit 0 = S_1, bit 1 = S_2; default and 3 = both; 0 is rejected); step 2 continues the sensors of step 1, and the status resource (Raw data 0x03) reports the mask of the current run. On the instrument the SETTINGS screen has three rows: "Zero cal both", "Zero cal S1" and "Zero cal S2". Pressing the row of the run in progress performs step 2; pressing another row after step 1 restarts with that row's sensors.

13.8 Differential zero (side-by-side), independent of the flip

Status: requirement and proposed routine; not implemented. The flip calibration above sets the *absolute* level of each sensor: it answers "what reading means level". It is needed once per sensor and is independent of the sensitivity (the zero is stored on the ratio r, ppm, which does not contain k, so a change of k moves neither the stored zero nor the level point). It is a separate question whether two sensors standing side by side on the same surface agree. Even directly after a flip they typically differ by a small amount (of the order of 5 µm), because each flip has its own noise and each flip measures the plate at a slightly different place. For the differential $S_1 - S_2$ this difference is a pure zero error and should be removable by a dedicated routine.

Routine. Place both instruments side by side on the same surface in the same orientation (the sign flips of Section 13.3 already make positive tilt the same direction for both). Average the readings r_1 and r_2 over a defined time (at least one minute; the differential has an Allan deviation of about 0.25 µm at 60 s, so 5 µm is well resolved) and form $e = \overline{r_1 - r_2}$. Each instrument absorbs half the error:

$$\text{zero}_1 \leftarrow \text{zero}_1 + \tfrac{e}{2}, \qquad \text{zero}_2 \leftarrow \text{zero}_2 - \tfrac{e}{2}, \tag{25}$$

so that afterwards $r_1 = r_2 = \frac{1}{2}(r_1 + r_2)$ before the correction. The mean $\frac{1}{2}(r_1 + r_2)$, i.e. the common level, is unchanged; only the difference is zeroed.

Independence. The two calibrations act on orthogonal combinations: the flip fixes each sensor's absolute level (the sum), the differential zero fixes only $r_1 - r_2$ and leaves the sum untouched. Neither needs a new constant: the differential zero is written into the existing per-sensor zeros (ppm of the ratio, like the flip result), so it is equally independent of k. It must be applied *after* a flip (a new flip overwrites the zero of that sensor and with it the difference), and repeated whenever either flip is repeated. The routine should refuse to run if the readings are unstable (for example a large pendulum amplitude, a doubtful flag set in many windows, or a large drift of $r_1 - r_2$ during the averaging), and use the same first-cycle discard and smoothing as the other averaged measurements.

Limits. The differential zero says that two instruments agree on *this* surface; it does not make the absolute level any more correct than the flips did, and it does not remove real differences

between the sensors (their own drift, the 3:1 pendulum amplitude ratio). If the two sensors differ in sensitivity, the difference grows with tilt, which a zero cannot remove: that is a gain matter, calibrated separately.

13.9 Precision measurement

A triggered mode reports one reliable number instead of a continuous stream (right knob on LIVE, API Commands 0x07). Since fw 0.10.71 it works as follows. After 81 contiguous batches (1.99 s) have arrived since the trigger, every new batch forms a new sliding window of the newest 81 batches, and the *first window that is clean* for both sensors is the result: the Hann-weighted means of S_1, of S_2 and of their difference over that window (the difference of the two means equals the mean of the per-batch differences). "Clean" is the window-level indicator of Section 13.5 (mean squared step of $\mathrm{Im}\,\underline{x}$ at most 6× the quiet floor, both sensors), so a disturbance has to have left the window completely before a value is accepted (and the quiet floor must exist, about 4 s after a start, so an early trigger can use up part of the 5 s) – "accept as soon as the last 2 s are clean". If no clean window exists 5 s after the trigger the measurement ends with an *error* and no value (the LIVE screen shows "Precision ERROR: disturbed", the API reports phase done with the error flag set and zeros). While waiting for calm the screen shows "Precision: disturbed" and the elapsed time. A dropped batch restarts the 2 s fill.

The design follows from the contiguous 19 h stream: the repeatability (between consecutive measurements, $S_1 - S_2$, nominal units) is 0.023 μm at night and 0.025 μm by day with this gate, against 0.078 μm by day without it; 1 % of the attempts fail and the mean duration is 2.02 s. The window shape hardly matters at 2 s or longer, and longer windows do not help: 5 s instead of 2 s gained about 20 % at night and nothing by day, because the daytime excess is rare disturbed windows (the largest step between neighbouring results was 8.7 μm without the gate and 0.25 μm with it), not averaging noise. The earlier scheme (average 64 batches that pass the per-batch flag, 4 s limit) is gone: it dropped individual batches, which breaks the cancellation of the 20 Hz resonance. **Status: implemented, not yet bench-tested.**

13.10 Outputs

The results are available over the API (measurements, the raw phasors of the latest batch as a topic, a bulk capture of raw ADC samples, and diagnostic counters), plus, since fw 0.10.65, a *continuous phasor stream* (Topic 0x05, resource 0x05) that pushes the phasors of *every* 64-cycle batch with a sequence number, so a host can record a gapless 40.7 Hz series and detect dropped batches. `Testing/2026-10-04_contiguous_phasor_capture/phasor_stream.py` logs it to CSV. The values are shown on the LIVE and diagnostic screens, and as the streaming and polling data used in the tests of this project. The diagnostic amplitudes (RMS, phase, and the "theoretical tilt" from the nominal 20 µV/(µm/m)) are computed directly from the phasors, independently of k, the attenuator and $\underline{A} - \underline{B}$.

14 Planned REV C Input Network and Excitation

Status: planned for the next hardware revision, not built. The goal is that all four ADC inputs have the same structure, hence the same phase, and that the mismatch and the drift between the A and B measurement paths are small, since those paths were found to dominate the zero drift (Section 12).

14.1 The network

Both paths use a 470 nF coupling capacitor, a resistive path to ground of 10 kΩ (same high-pass corner of 34 Hz for all four) and a 10 nF C0G capacitor at the ADC pin. In the A/B path the

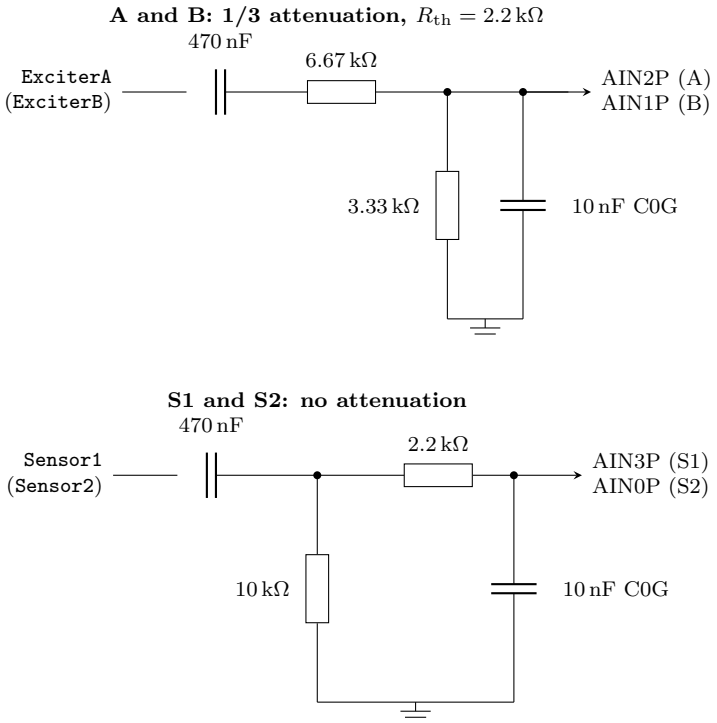


Figure 7: Planned REV C input networks. The 6.67 k/3.33 k pair is one matched resistor network (10 k total, ratio 2:1). All coupling capacitors are 470 nF, all pin capacitors 10 nF C0G.

resistor network is at the same time the ground reference, the 1/3 attenuator and the filter resistor: its Thévenin resistance is $6.67 \parallel 3.33 = 2.22\,\text{k}\Omega$, almost exactly the $2.2\,\text{k}\Omega$ series resistor of the S path. The S signals are not attenuated (attenuating a signal only to add gain again would waste resolution).

At f_0 (ADC input at gain 1 included):

Path	Gain	Delay	Peak at the pin	Remark
A and B	0.311	19.1°	0.88 V (74 % of full scale)	at 2.00 V RMS excitation
S_1, S_2	0.920	18.6°	29.9 µV per µm/m	PGA 16: ±2.5 mm/m

The delay difference between the S and the A/B path is 0.5°, so the phase constants of Section 9 reduce to the sensor's own delay (the extra 7 to 11° found in Section 9 comes from the old electronics in the sensor; its output is an op-amp output and therefore low impedance). The gain ratio between the paths is $k_{\text{phys}} = 16 \cdot 0.920/0.311 \approx 47$. The attenuation of the A/B path above f_0 is −68 dB at f_{MCLK}; the corner of the low-pass is 7.2 kHz.

14.2 Excitation chain

Status: proposal, nothing built or changed. The chain between the DDS and the two excitation outputs is proposed as follows (all values E12/E24):

Function	Value	Series
Coupling capacitor after the DAC	470 nF	E6
Ground reference (DC path)	6.8 kΩ	E12
First RC low-pass	680 Ω + 10 nF	E12
Second RC low-pass	6.8 kΩ + 1 nF	E12
Gain stage (OPA2191 or OPA2197)	R54 = 3 kΩ, R53 = 27 kΩ ($G = 10.0$)	E24
Inverter	10 kΩ / 10 kΩ	see below
Output isolation	22 Ω in series at each output	E24

Order of the stages. The DC block comes first, right after the DAC, so that the DAC's offset (344 mV) is never amplified. The two RC stages are passive and sit in front of the op-amp, whose input is high impedance (FET/CMOS type), so no buffer is needed. At f_0 the chain (including the DAC's 200 Ω source and the op-amp input) has a gain of 0.909 and a delay of 13.8°, a high-pass corner of 48 Hz, and attenuates by 97 dB at f_{MCLK}.

How critical is this filter? Not very. The ADC modulator runs at $f_{\mathrm{MCLK}}/2$, so the DAC's image at $f_{\mathrm{MCLK}} - f_0$ aliases exactly onto f_0 in the ADC and cannot be removed digitally. It is however tiny already: about −66 dB from the DAC's own sample-and-hold, a further −57 dB (relative to the passband) through the A/B network and about −66 dB from the modulator's input sampling (an estimate), together below −150 dB even without any exciter filter. The RC stages are for cleanliness (less RF into the op-amps and the sensor electronics), not for the measurement. Because the filter sits before the point where the excitation is split into A, B and the sensor supply, its gain and phase are common to all of them and cancel in the ratio. Precision is needed only after the split: the ratio of the inverter, and the A/B input networks (Section 14.1).

Why not an active filter. A Sallen-Key low-pass has a capacitor from the filter node to the op-amp output. Once the loop gain of the op-amp is gone at high frequency, the output resistance R_0 of the op-amp forms a divider with the input resistor through that capacitor, and the stopband floors at about R_0/R_1. For a 23 kHz unity-gain Butterworth stage (1 kΩ, 9.8 nF/4.9 nF) and a 10 MHz op-amp, the response at f_{MCLK} is −96 dB for an ideal op-amp, but −37 dB for $R_0 = 30\,\Omega$ and −27 dB for $R_0 = 100\,\Omega$. Two passive RC stages give −94 dB independent of the op-amp. The filter therefore stays passive and in front of the op-amp.

Gain stage and excitation level. With the chain above the excitation is $0.306 \cdot 0.909 \cdot G$. The nominal supply of the sensor is 2 V RMS (2.83 V peak); the A/B input network (Section 14.1) has a gain of 0.311.

R53 / R54	G	Excitation (peak / RMS)	A/B at the pin	Note
27k / 3k	10.00	2.78 V / 1.97 V	0.87 V (72 %)	OPA2197, preferred
13k / 1k5	9.67	2.69 V / 1.90 V	0.84 V (70 %)	OPA2197, lower impedance
130k / 15k	9.67	2.69 V / 1.90 V	0.84 V (70 %)	AD826 (see below)
150k / 15k	11.0	3.06 V / 2.16 V	0.95 V (79 %)	rail-to-rail part only

For a bipolar op-amp such as the AD826 (bias current 3.3 µA) the DC resistance seen by the non-inverting input ($6.8 + 0.68 + 6.8 = 14.3$ kΩ) must be matched by R53 ∥ R54; with 15k/130k (13.5 k) the offset is about −25 mV, with 10k/100k it would be −185 mV. A FET or CMOS input op-amp makes this irrelevant (offset in the microvolt range), which allows the lower resistances of the first two rows. The AD826 guarantees only about −2.8 V of swing on the −4.5 V rail, so 130k/15k works there with ≈ 80 mV of margin and 150k/15k does not.

The inverter ratio. $B = -A(1 + \varepsilon)$ with ε the ratio error of the two resistors. The static part scales the tilt by $\varepsilon/2$ and is calibrated away (±500 ppm for a matched network, up to ±1000 ppm for two 0.1 % resistors). Only the drift matters: $\Delta\varepsilon/2$ on the tilt scale is about 1 ppm/K for a

matched network and 5 to 12 ppm/K for discrete resistors, which is below the noise floor at any realistic tilt. If the B term were kept in the ratio (Section 12.8), the drift would act through $\underline{z}$ as $\Delta\varepsilon/4$, which would be 0.7 against 4 to 9 µm/m per kelvin at the present calibration. So the matched network pays off only if the B term is kept; with the B term dropped, two discrete 0.1 % resistors placed next to each other are sufficient, and a matched 1:1 network is worth using if it costs about a dollar or less. The finite loop gain of the op-amp adds a quadrature imbalance of $2f_0/\mathrm{GBW}$ (≈ 100 ppm for 50 MHz, about 500 ppm for 10 MHz, about 2400 ppm or 0.14° for the 2.2 MHz of the OPA2191), static and without consequence.

14.3 Components and tolerances

Function	Part	Note
Coupling capacitors (4)	470 nF X7R 50 V 0805, Samsung CL21B474KBFNNNE, LCSC C13967	Basic tier; library symbol `C0805_470n`
Pin capacitors (4)	10 nF C0G 50 V ±1 % 0805, Murata GRM2195C1H103FA01D, LCSC C441760	Extended tier (one-time fee); `C0805_10n_C0G`
A/B divider (2)	matched resistor network, 10 kΩ total, ratio 2:1, ratio tracking about 2 ppm/K	part to be chosen; one per channel, adjacent
S resistors (2 + 2)	10 kΩ and 2.2 kΩ, 0.1 % thin film recommended (1 % acceptable)	

Why C0G only at the pin. Sensitivity of gain and phase at f_0 to a 1 % change of the capacitor (from the corner of its own network):

Capacitor	Gain change per 1 %	Phase change per 1 %
10 nF at the pin (corner 7.2 kHz)	0.12 %	0.18°
100 nF coupling (corner 159 Hz)	0.004 %	0.007°
470 nF coupling (corner 34 Hz)	0.0002 %	0.0075°

The pin capacitor is about 700× more sensitive in gain and 24× in phase. Its temperature drift also matters most: a drift mismatch of only 0.5 % between the A and B capacitors would change the A/B ratio by about 600 ppm, the size of the drift observed on REV B (which uses capacitors of unknown, probably X7R, quality). The AC voltage across the coupling capacitor is only about 1.3 % of the signal, so the voltage coefficient of X7R is irrelevant there.

Simulated mismatch between two nominally identical channels (array ratio 0.1 %, absolute 0.1 %, discrete resistors 1 %, pin capacitor C0G 1 %, coupling capacitor X7R 10 %):

Pair	Phase (SD)	Gain (SD)
A vs B	0.16°	0.11 %
S_1 vs S_2	0.22°	0.21 %
A vs B, array absolute tolerance 1 %	0.23°	0.15 %
all pairs with X7R 10 % pin capacitors (for comparison)	1.5°	1 %

The expected drift of the A/B ratio between the two channels is about 3 ppm/K (array tracking about 2, capacitor about 1), against about 47 ppm/K on REV B (an estimate for REV C, a measurement for REV B).

14.4 Alternatives considered

- **Attenuating S_1 and S_2** like A and B (so that all four are identical): possible, and the PGA would give the full ±10 mm/m range, but it only attenuates in order to add gain again; the phase calibration handles the small difference anyway.

- **Capacitive divider** (2.2 k, then a series capacitor of 3.3 or 4.7 nF into the 10 nF at the pin): flat wide passband and light load, but the gain depends first-order on the capacitor ratio, which discrete C0G parts track to only about 10 to 15 ppm/K, against about 2 to 5 ppm/K for a matched resistor array. Not better on drift.
- **Analog multiplexers** at the S inputs (route each signal to either ADC channel and polarity): not adopted. Offsets are removed by the carrier and the demodulation, a gain drift only scales the tilt, and the errors found (the A/B balance, the slow drift of S_1) are upstream of any multiplexer at the ADC pins. The switches would add their own resistance, leakage and charge injection. A cheaper diagnostic is to swap the two sensor cables once and see whether the drift of S_1 follows the sensor or the channel.

14.5 Op-amp selection (REV C)

Status: shortlist verified from the TI datasheets, no part ordered. The AD826 of REV B was chosen for its output current, which the circuit does not need (a few mA suffice, and the supply current does not change when the sensors are attached). Requirements for the replacement: absolute maximum supply of at least 12 V (the rails are +5 V and −4.5 V, 9.5 V in total, which excludes the many parts limited to 5.5 V); output swing of at least ±3 V into 5 kΩ; input bias current of at most 1 nA (JFET or CMOS input); unity-gain stable; gain-bandwidth of about 2 MHz or more (first set to 5 MHz, relaxed after the loop-gain analysis below; a very fast part invites oscillation); low quiescent current (the instrument is battery powered); output current of at least 10 mA; dual in SOIC-8 with the same pinout as the AD826.

Part	LCSC	Stock	Price (10/100)	I_q/amp (typ/max)	GBW	Noise 1 kHz
OPA2191IDR (TI)	C543015	3239	$2.09 / $1.59	0.14 / 0.20 mA	2.2–2.5 MHz	15 nV/$\sqrt{\text{Hz}}$
OPA2197IDR (TI)	C139363	9501	$0.71 / $0.57	1.0 / 1.3 mA	10 MHz	5.5 nV/$\sqrt{\text{Hz}}$
OPA2192IDR (TI)	C110074	44738	$1.05 / $0.82	1.0 / 1.2 mA	10 MHz	5.5 nV/$\sqrt{\text{Hz}}$
OPA2172IDR (TI)	C545167	17	$1.17 / –	1.6 / 1.8 mA	10 MHz	7 nV/$\sqrt{\text{Hz}}$
OPA2156IDR (TI)	C1850241	4745	$2.02 / –	(not checked)	about 25 MHz	(not checked)
AD826ARZ (reference)	C652603	946	$5.19 / –	up to 7.5 mA	50 MHz	(not checked)

All are Extended tier (a single fee per board; from jlcsearch, snapshot 2026-10-01, not verified on the JLCPCB part pages). Common to the OPA2191, OPA2192 and OPA2197 (TI datasheets, SBOS701D, SBOS620E, SBOS737C): supply range 4.5 to 36 V; input bias ±5 pA typical (±20 pA maximum); rail-to-rail input range extending 0.1 V beyond both rails; open-loop gain of 120 dB or more at DC; unity-gain stable with a capacitive load of up to 1 nF directly on the output; standard dual pinout (OUT A, −IN A, +IN A, V−, +IN B, −IN B, OUT B, V+ on pins 1 to 8, confirmed in the OPA2191 datasheet). Differences:

	OPA2191	OPA2192 / OPA2197
I_q per package (2 amplifiers)	0.28 mA typ, 0.40 mA max	2.0 mA typ, 2.4–2.6 mA max
GBW (typical only)	2.2 MHz (±2.25 V) to 2.5 MHz (±18 V)	10 MHz
Slew rate	5.5–7.5 V/µs	20 V/µs
Output swing from the rail, 10 k / 2 k (typ)	15–50 mV / 60–200 mV	95 mV / 430 mV
Short-circuit current	±30 mA at ±2.25 V, ±65 mA at ±18 V	±65 mA
Open-loop output impedance at 1 MHz	700 Ω	375 Ω
Noise at 1 kHz	15 nV/$\sqrt{\text{Hz}}$	5.5 nV/$\sqrt{\text{Hz}}$
Offset (typ / max)	5 / 25 µV	5 / 25 (192), 25 / 100 (197) µV

None of the datasheets specifies values at the 9.5 V in use (the tables are at ±18 V and ±2.25 V). Offset and noise are irrelevant for the exciter (AC-coupled, and its noise cancels in the ratio).

Finite loop gain (single-pole estimate). Gain error and phase lag at f_0 from the closed-loop corner GBW/(noise gain):

Op-amp	Stage 1 ($G = 10$)	Stage 2 ($G = -1$)	Total phase lag
OPA2191, 2.2 MHz	$-0.007\,\%$, $-0.68°$	$-0.0003\,\%$, $-0.14°$	about $0.8°$
OPA2192 / OPA2197, 10 MHz	$-0.0003\,\%$, $-0.15°$	about 0, $-0.03°$	about $0.2°$

The amplitude error is negligible for all of them (below $0.01\,\%$), and the slew rate needed for 2.78 V at 2.6 kHz is only $0.045\,\mathrm{V/\mu s}$. What remains is a static phase lag, common to A, B and the sensor supply, which the phase calibration absorbs as long as the gain-bandwidth does not change with temperature (it is specified as typical only and was not verified). The inverter's lag appears as a static imbalance of B against A of $0.14°$ for the OPA2191. All three parts have a complementary input pair with a transition zone between about (V+) − 3 V and (V+) − 1.5 V (about +2 to +3.5 V here), where the CMRR drops; the inputs of the exciter stay near 0 V, in the good region, and must not be biased into that zone.

Recommendation. The OPA2191 is acceptable and is the preferred part where battery current has priority: 0.28 mA instead of 2 mA per package (1.7 mA less, about 16 mW from the 9.5 V rails), with the same 1 nF output stage, pA bias, high open-loop gain and a GBW above the 2 MHz floor. The OPA2197 is the fallback and the better choice if the supply current is not critical: it is 2.7× cheaper ($0.57 against $1.59 at 100 pieces), faster (4 to 5× the GBW, hence about 4× less phase lag), quieter and has the lower output impedance. The OPA2192 has the same speed as the OPA2197 with a tighter offset and the deepest stock (44 738), at a small premium for a precision that is not needed here. Red flags for the OPA2191: the GBW is typical only (no minimum, no temperature spread); the output drive at 9.5 V is not specified (±30 mA at ±2.25 V); the open-loop output impedance of 700 Ω calls for the isolation resistor; it is the most expensive and has the least stock of the three; some datasheet table rows (offset, drift, PSRR, the isolation-resistor table) were garbled in the text extraction and not relied on. The AD826 oscillation has not been root-caused, so the datasheet claims (including the 1 nF capacitive-load drive) must be confirmed on the bench with the real load, cable and isolation resistor, whichever part is chosen. Rejected: parts with a 5.5 V or 16 V supply limit, the TL072 (3 MHz), the OPA2140 (about 3.6 USD) and the OPA2134 (too much supply current).

14.6 Stability: compensation and layout (REV C)

Status: proposal. A small capacitor C_f across the feedback resistor (from the output to the *inverting* input) compensates the pole formed by the feedback resistance and the stray capacitance at the inverting pin, and restores the phase margin. (A capacitor from the output to the *non-inverting* input would be positive feedback and would make oscillation more likely.) It does not affect the signal: with R53 = 27 kΩ and 2.2 pF its pole is at 2.7 MHz, a thousand times above f_0. A usual estimate is $C_f \approx \sqrt{C_{\mathrm{in}}/(2\pi R_f\,\mathrm{GBW})}$:

Op-amp, feedback resistor	C_{in} (assumed)	C_f
AD826 (50 MHz), 100 kΩ	4 pF	0.4 pF
AD826, 27 kΩ	4 pF	0.7 pF
OPA2197 (10 MHz), 27 kΩ	5 pF	1.7 pF
OPA2197, 13 kΩ	5 pF	2.5 pF
OPA2191 (2.2 MHz), 27 kΩ	5 pF	3.7 pF
OPA2191, 13 kΩ	5 pF	5.3 pF

The stray capacitances are assumptions, so these are starting values to be tuned on the bench. Proposed for the board: pads for C_f across R53 and across R56 (the inverter's feedback), 0402 C0G, with 2.2 pF populated by default; 22 Ω to 47 Ω in series at each output to isolate cable capacitance; 100 nF plus 1 µF at each supply pin within a few millimetres; a short trace to the

inverting input, with no ground plane under that pin. The OPA2191 and OPA2197 are slower and more tolerant than the AD826, so these are mainly insurance (the OPA2191 needs the largest C_f, because of its lower GBW).

14.7 The supply-current problem on REV B

Observation. On both REV B boards the supply current is about 80 mA instead of the expected 20 mA; a thermal image shows that the excess is dissipated in the AD826. The current is essentially unchanged when the sensors are attached. The excess of about 60 mA is far above what two AD826 amplifiers should draw (at most about 15 mA together) and independent of the load, which points to an instability, most likely a high-frequency oscillation of one stage (a class-AB output stage that oscillates draws and dissipates exactly this kind of current), identical on both boards because it is a design property.

Probable cause (hypothesis). The feedback resistor R53 is 100 kΩ. With R54 10 kΩ the feedback network has 9.1 kΩ in parallel; with about 4 pF of stray capacitance at the inverting pin it forms a pole at 4.4 MHz, right at the crossover frequency of the loop (GBW/G = 4.5 MHz), and costs about 46° of phase margin.

R53 / R54	Output DC (3.3 µA)	Feedback pole	Phase margin lost
100k / 10k (now)	−280 mV	4.4 MHz	about 46°
47k / 4k7	−460 mV	9.3 MHz	about 26°
22k / 2k2	−540 mV	20 MHz	about 13°
150k / 15k	−120 mV	2.9 MHz	about 57°

This shows why the resistors cannot simply be lowered on the AD826: lowering R53 removes the term of the bias-current compensation, and the output offset gets worse (to −0.54 V for 22k/2k2, which would clip the negative peaks).

Diagnosis (to do, first). Look for oscillation with the oscilloscope before changing anything. Probe the outputs (U16 pin 1 ExciterA and pin 7 ExciterB) and the supply pins (pin 8 and pin 4, at the decoupling capacitors C67, C68, C73, C74); the inverting input (pin 2) should not be probed, because the probe capacitance alone can start or stop an oscillation. Use a ×10 probe at full bandwidth (20 MHz limit off) with a spring ground tip, AC coupling, 20 to 50 mV/div, 100 ns/div to see the HF structure and 100 µs/div to see where on the 2.6 kHz cycle it occurs (peak-detect or persistence), or an FFT of 0 to 50 MHz. Typical signs are a thick trace, an HF burst of 1 to 30 MHz on the peaks or the zero crossing, and HF ripple on the supply pins. Two extra checks: whether the extra current stays with the DDS stopped (then the oscillation is self-sustaining), and how the current splits between the +5 V and the −4.5 V rail. If the outputs and rails are clean and the current stays high, the cause is elsewhere (for example the −4.5 V rail), and the resistor changes below should not be made.

14.8 REV B rework options (not applied)

Status: analysis only, waiting for the oscilloscope check. Only resistor values can be changed on the existing boards, and the AD826 stays. The feedback resistors and the DC resistance in front of the non-inverting input must then be changed together: a low feedback impedance for stability needs a low DC resistance R50 + R51 (otherwise the bias current offset grows). Candidates (all E24, gain 11.0, R52 stays at 0 Ω):

R50, R51 / R53, R54	Exc. RMS	DC offset	Pole	PM lost	HP / LP
10k, 6k8 / 100k, 10k (now)	2.24 V	−280 mV	4.4 MHz	46°	156 Hz / 21 kHz
1k5, 1k / 27k, 2k7	1.95 V	−2 mV	16 MHz	16°	936 Hz / 145 kHz
2k2, 1k / 36k, 3k6	2.09 V	+3 mV	12 MHz	20°	663 Hz / 145 kHz
2k2, 1k5 / 43k, 4k3	2.08 V	+8 mV	10 MHz	24°	663 Hz / 96 kHz

The first candidate (R50 = 1.5 kΩ, R51 = 1 kΩ, R53 = 27 kΩ, R54 = 2.7 kΩ) gives a negligible output offset, about 16° of phase-margin loss instead of 46°, an excitation of 1.95 V RMS (close to the sensor's nominal 2 V) and a load of 30 kΩ. The prices are a higher high-pass corner (936 Hz: gain 0.82, phase lead 18.5° at f_0) and a weaker filter (−34 dB instead of −48 dB at f_{MCLK}), both common to A, B and the sensor supply and therefore harmless, as argued above. The phase-margin figures depend on the assumed 4 pF; the real test is the supply current after the change, which should fall towards 20 to 25 mA if instability is the cause. Where a small capacitor across R53 can be fitted (Section 14.6), it is the least invasive first step: it leaves the DC offset at −0.28 V.

15 Status, Open Items and Proposals

15.1 Hardware deviations from the schematic (REV B)

Part	Schematic / manufactured	Working boards
R52	68 kΩ	0 Ω (removes the DC offset from the op-amp bias current)
R46, R47 (sensor filter)	680 kΩ (mistake)	680 Ω
R48, R49 (A/B filter)	680 kΩ (mistake)	680 Ω

15.2 Planned changes for REV C (Section 14)

Identical input structure for all four channels with a 470 nF coupling capacitor and a 10 nF C0G pin capacitor each; A/B through a matched 2:1 resistor network (1/3, 2.2 kΩ), S1/S2 with 10 kΩ to ground and 2.2 kΩ in series; a new excitation chain (passive filter in front of a low-bias op-amp, OPA2191 (battery current) or OPA2197 (cost, speed), R53/R54 = 27k/3k, excitation about 1.97 V RMS) with compensation pads and output isolation (Sections 14.2, 14.5, 14.6); R46–R49 and R52 as the corrected values of Section 15.1 become the schematic values. Library symbols `C0805_10n_C0G` (C441760) and `C0805_470n` (C13967) have been added. The excess supply current of the REV B excitation driver (Section 14.7) is to be diagnosed first; resistor rework options are analysed in Section 14.8 but not applied.

15.3 Implemented and verified

Excitation, filters and input networks as described; simultaneous 24-bit sampling locked to the excitation; single-bin demodulation over 64 periods (per-position integer accumulation, fw 0.10.64); the projection estimator $\mathrm{Re}[\underline{S}/(k(\underline{A}-\underline{B}))]$ (fw 0.10.62) with zero calibration and precision measurement; continuous phasor stream (fw 0.10.65); Hann display stream with doubtful-reading flag and sliding-window precision measurement (fw 0.10.71, not yet bench-tested); cleanup in fw 0.10.72: per-batch quality flag, moving average and bulk phasor log removed.

15.4 Findings that need action

Item	Detail
Diagnostic amplitude scale (Sec. 7)	**Fixed 2026-10-06:** `phasor_peak_mv` used $2400/\mathrm{PGA}/2^{23}$, twice the datasheet LSB $2.4\,\mathrm{V}/(\mathrm{PGA}\cdot 2^{24})$. Sensitivities calibrated against the old diagnostic values must be redone (k replaced them in fw 0.10.69).
k (Sec. 12.5)	k is one empirical sensitivity per sensor, stated for PGA 1 (default 0.0213 per mm/m, Section 13.3; the attenuator, gain, d_0 and sensitivity constants were merged into it by fw 0.10.69) and the phase is a separate constant. The default is derived from the nominal sensor sensitivity and the network gains (Section 13.3); a calibration against a known tilt replaces it.
Zero calibration (Sec. 13.7)	**To redo.** $\lambda = 0$ (drop the $-\underline{B}$ term) has been in the firmware since 0.10.62; the stored zero offsets still contain about 3.6 mm/m of the old balance term. Verify λ with a deliberate imbalance (Sec. 12.7, step 3).
Differential zero (Sec. 13.8)	Requirement: a routine that zeroes $S_1 - S_2$ side by side (each sensor takes half the error, e.g. the observed $\approx 5\,\mu\mathrm{m}$), independent of the flip calibration and of the gain. Not implemented.
Phase calibration (Sec. 9)	**Implemented in fw 0.10.66** (approach 2), with estimated defaults ($-6.0°$, $-9.15°$), proper calibration still to do: run-time reference $\underline{D} = \underline{A} - \underline{B}$ at 90° ($A \approx 90.5°$, $B \approx 269.5°$, confirmed in three runs), S_1/S_2 near 90° or 270°; only two EEPROM constants δ_1, δ_2 (phase of k; estimates about $-10°$ and -10.6 to $-14.3°$, to be determined properly from a tilt change). The absolute phase is *not* reproducible (122.85, 122.67, 334.39°; Sec. 10.5), which is why a fixed absolute constant was rejected.
Sensor phase and impedance (Sec. 9)	After the networks, the sensors add about $+11°$ (S_1) and $+7°$ (S_2) of delay (old electronics in the sensor; the output is a low-impedance op-amp output, not measured).
B relative to A (Sec. 9)	179.0° instead of 180°, unexplained but stable in three runs (178.94, 178.96, 179.00°).
Pendulum resonance (Sec. 13.5)	About 20 Hz, at the batch Nyquist frequency; dominates the sensor noise. The LIVE display and the precision measurement now use Hann windows over contiguous batches (fw 0.10.71, bench test pending); the 8-batch boxcar of the API value is unchanged; individual batches must not be dropped.
First cycle after start (Sec. 13.7)	The first cycle after `start()` can be corrupt on all four channels (seen in bulk captures); firmware does not yet discard it.
Harmonics on S_1/S_2 (Sec. 11.4)	2nd and 3rd harmonics of $-21/-15$ dBc (S_2) are rejected exactly, the 7th/9th fold onto the carrier and are unchecked.
$\lvert\underline{D}\rvert$ drift (Sec. 12.7)	About 160 ppm in 74 min (906 ppm in 17.5 h): excitation or path drift? A constant-tilt test is needed.
Sensitivity k (Sec. 13.3)	Default 0.0213 per mm/m at PGA 1 (nominal). The 2026-09-29 calibrations correspond to about 0.013, i.e. the sensors give about 60 % of the nominal signal; cause open. To be recalibrated against a known tilt (also because of the diagnostic scale fix and the phase rotation).
Excitation DC and headroom (Sec. 6)	Residual offset about -0.3 V on ExciterA and headroom of about 3.5 V against the AD826's ± 3.3 to ± 3.8 V swing; calculated, not yet measured on the bench.
Excess supply current (Sec. 14.7)	Both REV B boards draw about 80 mA instead of 20 mA; the AD826 is hot, and the current does not change with the sensors attached. Probable cause: HF oscillation of the gain stage (feedback pole of the 100 kΩ R53 at the crossover). To do: oscilloscope check of the outputs and supply pins (Sec. 14.7) before any rework (Sec. 14.8).
Sunlight / long-term (Sec. 12.6)	The bump at about 21 h of the 24 h run is very likely sunlight on the sensors (a real input). The clean window for the drift analysis is about 17.5 h.

15.5 Suggested tests

- A deliberate, known imbalance of the excitation (for example a resistor across R56) to measure λ and the complex k directly.
- Repeated forced-charge steps on an undisturbed plate, to obtain several clean steps of $\underline{z}$.
- A second temperature sensor at the B input network, to separate thermal gradients between the A and B paths from other causes.
- The same event-based correction on the 1 h run at PGA 1 as an independent check.
- A constant-tilt, constant-temperature run with the phasor stream (fw 0.10.65) to see whether $|\underline{D}|$ and the sensor phasors drift together (excitation) or apart (path).
- A deliberate tilt step (not the almost flat zero calibration) with the stream to measure δ_1, δ_2 from the change of $\underline{S}$.
- Post-processing of the full 24 h phasor stream when it is available: smoothing windows, Allan deviation, and the 7th/9th-harmonic question.